

Supplementary Material — meryemAircraft

Supplementary material to "The Architectural Cost of Hybrid VTOL: meryemAircraft, a Propeller-Driven Tail-Sitting Blended-Wing-Body Without a Dedicated Lift System".

These six files carry the derivations behind the results stated in the paper. Each was a section of an earlier, longer version and is reproduced without abridgement, so that every number quoted in the main text can be traced to the calculation that produced it. The computational setup, the scripts, and a running record of the corrections made during the study are in the repository the paper cites.

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Supplementary S1 — Independent checks on the two assumed aerodynamic coefficients

Supplementary material to "The Architectural Cost of Hybrid VTOL: meryemAircraft, a Propeller-Driven Tail-Sitting Blended-Wing-Body Without a Dedicated Lift System".

This material is Section 3.10 of the paper, reproduced here in full. It is reproduced here in full so that every number quoted in the main text can be traced to the calculation that produced it. The computational setup, the scripts, and the record of what failed along the way are in the repository the paper cites.

S1.1 Independent checks on the two assumed coefficients

The two coefficients that carry the most weight in Section 3 — span efficiency and zero-lift drag — were assumed rather than derived. They remain assumed. What follows does not replace them with computed values; it asks a narrower question that can be answered honestly: **are the assumed values inside the range that a calculation gives, and on which side?**

What the vortex-lattice method is being asked for, and what it is not. Four results in this paper come from a vortex-lattice solution: the span efficiency below, the neutral point

of Section 3.17, the twist required to trim, and the roll damping of Section 2.10. Falkner's account of the method's accuracy [24] separates the quantities it settles quickly from those it does not, and the separation is favourable here. He reports that "the grading of spanwise circulation converges quickly" and that "the local aerodynamic centre can be adequately defined by the use of two chordwise terms only", while "a reasonably accurate calculation of pressure distribution in the neighbourhood of a discontinuity in plan would require at least three or four terms". Every quantity taken from the method here is of the first kind — circulation and aerodynamic centre — and none is of the second. That is consistent with what the convergence study found: the neutral point moved 0.26 percent over a threefold refinement, while the span efficiency needed four times the default resolution before it stopped returning values above unity, which is physically impossible for a planar wing and is the signature of an under-resolved plan discontinuity — this planform is cropped, and therefore has one.

A second boundary is worth stating because it marks where the method is not used. For wings of low aspect ratio and high sweep, leading-edge vortex separation dominates and the linear method must be extended by a suction analogy to follow it [25]. That regime is not this wing's cruise condition — aspect ratio 6.03 at 45° root sweep, at incidences under eleven degrees — but it *is* the regime of the transition incidences in Section 3. No vortex-lattice result is quoted there, and this is why.

A third boundary was found by reading, and it is the sharpest of the three because it is a measured number rather than a statement of scope. A study of a tactical blended-wing-body UAV compared a vortex-lattice solution against RANS at the same condition and found the vortex-lattice lift coefficient low by **thirty to thirty-eight percent** across incidences from -4° to 12° , and excluded the method from its trim analysis on that basis [39]. That is a configuration of the same class as this one, and the four results above rest on the same kind of solution.

Two things keep the argument of the previous paragraphs standing, and one of them cannot be checked. The deviation is nearly constant with incidence — 37.8, 37.7, 38.2, 37.2 and 30.7 percent — which is the signature of a multiplicative error in the magnitude of the loading rather than an error in its distribution. A neutral point is a ratio of derivatives and a static margin is a ratio of lengths; a factor common to lift and moment cancels in both, and the twist effectiveness would be *under*-stated, meaning the twist actually required to trim would be smaller than Section 3.17 reports rather than larger. That argument depends entirely on the moment scaling with the lift, and the pitching-moment comparison in that paper is published with its values withheld. **The check that would settle it is precisely the one the source does not permit.**

What can be said without it is narrower. The likeliest origin of a deficit that large is the volume of the centre body, which a vortex-lattice method represents as a camber surface of zero thickness and which on a blended wing carries a real share of the lift; the cited study does not say whether its model included the body at all. The planform solved here carries its centre section as part of the lifting surface, so the same deficit should not transfer at full size. But it does not transfer to zero either, and **the honest statement is that the vortex-lattice results in this paper carry an untested magnitude error of**

unknown size, bounded above by a published comparison on a similar configuration, and that every use made of them here is of a kind that a magnitude error *would not* disturb if the moment scales with the lift by the same factor. That conditional cannot be discharged from the source, and nothing else in the literature read here discharges it either. It is stated as a conditional and carried as one. Confirming it would take a viscous or Reynolds-averaged solution of this planform, which Section 4 names as the first thing worth computing next.

Span efficiency. A vortex-lattice solution of the planform of Section 2.8 [15] gives an inviscid span efficiency of 0.99. That is not the same quantity as the 0.85 used here. The vortex-lattice figure counts only the departure of the induced drag from the elliptic ideal; the 0.85 is an Oswald-type efficiency that also carries the viscous drag due to lift, which for a clean wing runs at roughly 85 to 90 percent of the inviscid value. The two are consistent. Reporting the calculation as an improvement on the assumption would be a category error, and it is not claimed.

Section 3.17 unsettles this, and the ratio has since been computed rather than assumed. The 0.99 belongs to the untwisted planform, and the untwisted planform cannot be trimmed. An earlier version of this paper carried the trimmed case forward with the same 85-to-90-percent rule and reported an implied Oswald value of 0.735 to 0.78. That rule was borrowed, and it turns out to be too pessimistic.

The calculation replacing it is the one the method of Section 3.10 was already equipped to make. The vortex-lattice solution gives the loading of the twisted wing station by station; the section solver used for the zero-lift drag build-up is then called at **each station's own local lift coefficient** rather than at zero lift, and the profile drag integrated across the span. This is the two-dimensional-viscous-coupled-to-three-dimensional-circulation construction used in the non-linear vortex-lattice literature [40], carried out on this planform. Before any result is taken from it, the strip decomposition is checked against the solver it comes from: summing the strip loads reproduces the solver's own lift coefficient to six decimal places, and the strip widths sum to the span.

Taking the lift-dependent drag to be the induced drag plus the rise in profile drag above its zero-lift value — the quantity an Oswald efficiency has to carry, since the zero-lift part is already inside C_{D0} :

	Inviscid e	Oswald e	Ratio
Untwisted planform	0.990	0.931	0.940
Trimmed, -9° washout	0.859	0.817	0.951

The borrowed ratio was wrong in the favourable direction and the conclusion is unchanged in the unfavourable one. The viscous penalty is 5 to 6 percent rather than 10 to 15, but the trimmed wing starts from 0.859, so the Oswald efficiency lands at **0.817 — below the assumed 0.85 by 3.9 percent**, not at the 0.735 to 0.78 previously feared. At that value the cruise lift-to-drag ratio is **11.87 against the 12.04 the assumption gives**, a shortfall of 1.4 percent rather than the 3 to 5 percent Section 3.17 had allowed for.

Two limits belong with the number. The vortex-lattice sections are symmetric, so the profile drag is that of a wing reaching each local lift coefficient without camber; a cambered section reaches the same lift at lower incidence and usually at lower drag, which makes this a **lower bound on the efficiency** rather than an estimate of it. And the strip method ignores sweep. Evaluating the same integral under simple-sweep theory instead — normal-component velocity and chord throughout — halves the profile drag, which is not a measure of uncertainty but a sign that the transformation does not apply to friction: the pressure field is set by the component normal to the sweep line, but the boundary layer runs over the surface at the full freestream speed. The flow-aligned convention is used here, and it is also the convention the zero-lift build-up uses, so the two numbers compose.

The same solution gives a lift-curve slope of 3.87 rad^{-1} against the 4.72 rad^{-1} that the transition simulation of Section 3.15 assumes — eighteen percent lower, and in the unfavourable direction. Supplementary S5.6 reports what that does to the transition results.

Zero-lift drag. A strip calculation over the span, taking section drag coefficients at zero lift from a physics-informed aerofoil model [16] and adding the tip frames and the propeller hubs, gives:

Contribution	Lower bound	Upper bound
Wing and body	0.0073 (clean surface)	0.0129 (transition tripped)
Tip frames, faired	0.0043	0.0043
Tip-propeller hubs	0.0015 (30 mm can)	0.0020 (50 mm can)
Subtotal	0.0131	0.0191
Excrescence allowance	none	+10 %
Total	0.0131	0.0210

The two columns are deliberately not the same calculation. The lower bound takes the clean surface, the smaller hub and no allowance for excrescences; the upper bound takes the tripped surface, the larger hub and ten percent for fasteners, joints, antennas and surface imperfection. The interval is a bracket rather than an error bar, and the allowance row was omitted from an earlier version of this table, which therefore did not sum to its own total.

The frame term reproduces the 0.0043 of Section 3.3, which was reached by a different route, and it comes out the same for the heavy design — an independent confirmation of the scale invariance claimed in Section 3.9. The hub term is bounded by hardware rather than guessed: each tip rotor must deliver about 0.83 kgf on a 0.20 m propeller, which places it in the standard 22 mm stator class whose outer cans measure roughly 28 mm.

The assumed 0.0248 lies above the whole of that range. The assumption is therefore conservative in every scenario considered, not merely in the pessimistic one. Because range is linear in lift-to-drag ratio, the reference designs of Sections 3.7 and 3.8 would gain rather than lose if the calculation were adopted — which is the reason it is not

adopted. An assumption that is declared and shown to be conservative is a smaller target than a calculation whose weakest link, discussed in Supplementary S5.4, is the treatment of a twenty-five percent thick centre body as a two-dimensional section.

A three-dimensional solution for the centre body. The weakness just named has since been removed. Supplementary S5.17 lists it first among the places these results should be attacked, and the calculation it asks for has now been carried out: a structured Reynolds-averaged solution over the planform of Section 2.8, at the cruise Reynolds number and at zero lift, resolving the wing and blended body as a three-dimensional surface rather than as stacked sections. It gives a wing-and-body zero-lift drag of

0.01475 with the Spalart-Allmaras closure and 0.01201 - 0.01253 with $k-\omega$ SST — a range rather than a single figure, and not for one reason but for two. Nothing in these solutions selects between the two closures, so the eighteen percent between them is carried openly; and the SST value itself is not unique, because two well-converged solutions of the same case from two materially different starting fields settle four percent apart. That second finding is set out below, since it bears on how much weight any of these numbers will hold.

Against the 0.0129 of the tripped strip estimate, the Spalart-Allmaras value is fifteen percent higher and the SST values between two and seven percent lower; the strip method is therefore bracketed rather than simply beaten. Substituting each in turn for the wing-and-body row, and carrying the larger hub and the ten percent allowance in every case so that the three are compared on one convention, gives **0.0201 with the low SST value, 0.0207 with the high one and 0.0231 with Spalart-Allmaras. The assumed 0.0248 lies above all of them**, so the conclusion of the previous paragraph survives the more expensive calculation under either closure and under either initialisation; the margin is twenty-three percent on the most optimistic value and seven percent on the conservative one. The assumption is not replaced here either, for the same reason as before. Where a single number is needed downstream, the conservative value is carried.

Neither range is a probabilistic uncertainty band, and neither is reported as one. The remaining terms are measured rather than asserted:

Source of uncertainty	Magnitude
Iterative convergence	0.1 %
Spatial discretisation, three grids of 0.27 - 2.66 M cells	< 0.1 %
Wall resolution, $y^+ 43 \rightarrow 20$	10 %
Wall resolution, $y^+ 20 \rightarrow 1$, Spalart-Allmaras	+ 1.6 %
Wall resolution, $y^+ 20 \rightarrow 1$, $k-\omega$ SST	– 6.7 %
Turbulence-model spread, $k-\omega$ SST against Spalart-Allmaras, at $y^+ \approx 20$	8 %
Turbulence-model spread, the same pair at $y^+ \approx 1$	18 %
Initialisation spread, $k-\omega$ SST at $y^+ \approx 1$, two starting fields	4.3 %

The dominant term is the turbulence model, not the grid — which is worth stating plainly, because grid convergence is the check a reader expects and it turns out to bound the smallest of the terms. The last two rows are labelled *spread* rather than *uncertainty* deliberately: two closures do not sample a distribution, and the interval between them carries no claim that the true value lies inside it. The same caution applies to the last row, which reports two starting fields and not a population of them. Three further results are recorded because they are easy to get wrong in either direction.

The forces converge far more slowly than the residuals: at a velocity residual of 1.6×10^{-5} the computed drag was still fifty-five percent above its converged value, so a solution stopped on residuals alone would have been badly wrong.

The wall-resolution sensitivity does not saturate, and it does not even share a sign between the two models. Resolving the wall raises the Spalart-Allmaras drag by 1.6 percent and lowers the $k-\omega$ SST drag by 6.7 percent, so the spread between the models doubles — from 8 percent at $y^+ \approx 20$ to 18 percent at $y^+ \approx 1$. No single wall-resolution correction is therefore assumed. An earlier version of this section quoted a single value with a five percent band, obtained by averaging one model at $y^+ \approx 1$ against the other at $y^+ \approx 20$; that mixes two wall resolutions and understates the model-form term.

The wall-resolved SST solution is not independent of its starting field. It cannot be started from a uniform field at all: from uniform initial conditions the run develops a localised region of non-physical turbulent kinetic energy near the leading edge at mid-span, which decays over some two thousand iterations without reaching a physical level and then diverges. Two starts that do converge were therefore compared — one warmed from the converged Spalart-Allmaras field, one mapped from the converged SST solution at $y^+ \approx 20$ on the coarser grid. The two initial fields differ by an order of magnitude in peak eddy viscosity and by a factor of six in peak turbulent kinetic energy; the cases are otherwise identical, sharing the same grid file, the same transport and turbulence properties, and the same boundary conditions. Both ran five thousand iterations without bounding, the mapped start reaching the lower velocity residual of the two, 1.0×10^{-7} .

They do not agree. The drag coefficients are 0.01253 and 0.01201, four and three tenths of a percent apart, and the difference lies almost entirely in the pressure component: 0.00386 against 0.00336, thirteen percent, while the viscous components agree to three parts in a thousand. A converged steady solution should not depend on its starting point; here it demonstrably does.

The two solutions are not, however, of equal standing, and the criterion that separates them is geometric rather than numerical. The sections are symmetric and untwisted and the incidence is zero, so the lift coefficient must vanish. The mapped solution returns 1.3×10^{-4} and the warm-started one 1.5×10^{-3} — an order of magnitude further from the value the geometry requires. The same asymmetry appears locally: at mid-span and three-quarter chord the upper and lower surface pressure coefficients differ by 0.0011 in the mapped solution and by 0.0249 in the warm-started one, where symmetry requires zero, and the ratio holds at every station examined. The induced drag of that residual lift is negligible — of order 10^{-7} , nine parts per million of the total — so the asymmetry is a

symptom and not the cause of the drag difference; but it is a physical test that one solution passes and the other does not.

Two further diagnostics bear on where the difference lives. The reverse-flow area on the wall is 0.02 percent in both solutions and occupies the same streamwise interval in both, so the two are not settling into visibly different separation topologies. And the pressure-drag difference is distributed almost evenly across the inner span — each of eight bands carries roughly an eighth of it — and concentrated in the rear quarter of the chord, where the mapped solution recovers more pressure. An earlier version of this section read the even spanwise distribution as evidence against a second solution branch; that inference is withdrawn, since a second stationary state need not be spatially localised. What can be said is narrower: the two solutions share a separation topology, and the one that better satisfies the symmetry the geometry imposes also recovers more trailing-edge pressure.

Both values are reported, because a third starting field has not been tried and nothing shows one would fall inside the interval; 0.01201 – 0.01253 is a computed spread and not a bound. Where a single value from this pair is wanted, the mapped solution is taken as the reference state, on the grounds that it carries the smaller residual lift and surface asymmetry. That is a selection criterion and not a proof: it does not establish that the warm-started solution is unphysical, only that it is further from a symmetry the geometry requires. The defence that a non-symmetric mesh would bias both cases equally is a supporting argument rather than a demonstration, since the equations are non-linear and a fixed mesh bias can couple differently to two different starting fields.

What this does not settle. The solution is fully turbulent throughout. It therefore speaks to the tripped row of the table above and not to the clean-surface row, and the gap between those two rows — 0.0073 against 0.0120–0.0148 — is now the largest single uncertainty in the zero-lift drag, larger than the spread between the two closures. Closing it requires a transition-sensitive model, which needs a wall-resolved grid and a two-equation formulation at the same time. That combination now converges, and the Langtry-Menter γ - $\text{Re}\theta$ model reproduces the ERCOFTAC T3A flat plate to within a few percent in the turbulent region while predicting transition about twenty-five percent early in Reynolds number. But T3A is a bypass case at three percent freestream turbulence, and the present cruise condition is an order of magnitude quieter; attempts to validate the model in that regime were not successful, for reasons traced to leading-edge turbulence production and freestream decay rather than to the model itself. The clean-surface bound therefore remains where the strip calculation left it, and no transition-model drag figure is claimed.

Internal volume. One closure that the mass budget does not address is whether the payload fits. The body's gross internal volume follows from the thickness distribution: 185 litres for the light design. Taking rather more than half of that as usable and subtracting fuel, buffer, powerplant and avionics leaves about 80 litres for 13 kg of payload, which requires a mean density of only 0.16 kg per litre. The configuration is mass-limited rather than volume-limited, and the choice of mission therefore constrains the structure and the load paths rather than the internal arrangement.

Supplementary S2 — A component build-up of the mass budget

Supplementary material to "The Architectural Cost of Hybrid VTOL: meryemAircraft, a Propeller-Driven Tail-Sitting Blended-Wing-Body Without a Dedicated Lift System".

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S2.1 A component build-up of the mass budget

The fractions used in Sections 3.7 and 3.8 are asserted, and Section 4.4 says so. This section replaces the assertion for the light design with a build-up from components. The rule followed throughout is that no item may be derived from the fraction it is meant to test: every line comes either from the geometry and a stress calculation, or from a specific quantity — an areal density, a specific power — stated openly and then varied.

Structure. The wetted area follows from the planform of Section 2.8 and the NACA 00xx thickness distribution: 4.14 m² against 1.98 m² of planform. A carbon-epoxy sandwich shell at 1.5 kg m⁻² gives 6.20 kg, with ribs, bulkheads and bonded joints taken at 45 percent of the shell. The tip frames are sized by a vertical landing case, since this aircraft lands on them: a 3 g arrival, half the weight through one frame, the post treated as a cantilever of the stated length, giving 0.95 kg for both frames including fittings. That case is not shown to be the worst one — an off-axis touchdown, a ground gust against the planform standing on its tail, or the thrust moment of the nose pair may govern the frame root or the joint into the wing instead — and no combined case was run. Fasteners, adhesive, filler and paint are charged at 10 percent of primary structure and access panels at 6 percent. The total is 11.88 kg, 23.8 percent of take-off mass.

Span bending is not what sizes the spar, and this is worth recording, because a thick blended centre body invites the assumption that it must be. At an ultimate load factor of 5.25 the root bending moment is 934 N·m; carried at 400 MPa over a structural depth of 0.9 times the root thickness, the caps require 10.7 mm² of carbon and weigh 41 grams. The claim that figure supports is narrow and is stated narrowly: **global span bending is not the sizing driver in this static model**. It says nothing about the failure modes the model does not contain — sandwich and face-sheet buckling, core shear, torsion, load introduction at the frame roots and the nose mount, minimum manufacturing gauge, damage tolerance, aeroelastic margin — and none of those is a reason to reduce the shell and internal-structure allowances above. Those allowances are where such mass would have to live, and the 41 grams does not license trimming them.

Propulsion. The nose motor is sized by hover peak power and the engine by cruise, which is the configuration's central claim and is visible in the budget as such: 2.73 kg of electric machine against 2.60 kg of engine and generator, at 4 kW kg⁻¹ and 1 kW kg⁻¹ respectively.

Adding the tip motors at twelve percent of nose power, the propellers, the coaxial hubs, the power electronics at 20 kW kg^{-1} , the engine mounting with its cooling and exhaust, and the power cabling gives 7.60 kg , 15.2 percent.

Systems, energy and contingency. Avionics, fuel system, strip actuation, signal harness and payload interface total 2.84 kg . There are no elevon or rudder actuators to add, since the aircraft has neither; the strip is its only moving aerodynamic device and its actuation is carried here. Fuel and battery are as sized, 9.80 kg . A contingency of 12 percent of dry mass — ordinary preliminary-design practice — adds 2.68 kg .

The battery buffer is specified by power, not by energy, and this has not been stated before. It must supply the difference between what the propellers demand and what the engine delivers — but that difference has to be taken at one station, and an earlier version of this section did not take it at one. It subtracted the engine's 2.6 kW of *shaft* power from the rotor's 10.9 kW of *shaft* power and divided the remainder by the buffer mass, giving 4.61 kW kg^{-1} . The buffer sits on the electrical bus, which is neither of those stations. Running the chain of Section 2.12 link by link instead:

Station	Light design, hover
Nose propeller shaft	10.90 kW
÷ electric machine, 0.92	11.85 kW
÷ power electronics, 0.95	12.47 kW demanded at the bus
Engine shaft $2.60 \text{ kW} \times$ generator 0.90	2.34 kW supplied at the bus
Buffer	10.13 kW

which is **5.63 kW kg^{-1}** from 1.8 kg , or about 31 C at 180 Wh kg^{-1} — twenty-two percent above the figure previously carried, and in the unfavourable direction. Taking the tip pairs as well, which Section 3.15 shows the aircraft needs in order to leave the ground at all, the bus demand rises to 14.00 kW and the buffer to 11.66 kW , or **6.48 kW kg^{-1}** at 36 C . The heavy design sits on the same line: 198.5 kW of buffer on 40 kg , **4.96 kW kg^{-1}** , rising to 5.54 kW kg^{-1} if the tip pairs are counted. Energy is not the binding constraint until roughly 140 seconds of hover, well beyond the profile of Section 3; below that the buffer is power-limited. That places the buffer well past the high-power lithium-ion or lithium-polymer class — cells rated for twenty to thirty times their capacity in continuous discharge — and the paragraphs below measure how far past.

How demanding, measured against the only figures this study has read, is the most exposed number in the whole mass budget. Bacchini and Cestino, sizing electric VTOL aircraft, take 735 W kg^{-1} at pack level from an automotive traction pack and report that "Li-ion batteries for power applications have... specific power from 700 to 1300 W kg^{-1} " [21]. The buffer here asks for **5.63 kW kg^{-1}** to hover and 6.48 to take off, four and a third to nearly nine times the top of that range. Sized at figures from that range instead, on the hover requirement alone:

Buffer specific power	Buffer mass	Fraction of MTOW
0.735 kW kg ⁻¹ (the pack they assume)	13.8 kg	27.6 %
1.30 kW kg ⁻¹ (top of their quoted range)	7.8 kg	15.6 %
5.63 kW kg⁻¹ (this study, implicitly)	1.8 kg	3.6 %

At the top of their range the buffer would be 6.0 kg heavier than budgeted, against the 2.2 kg of unallocated mass this section leaves. **The light design's mass budget would not close.**

The defence available at that point was that those figures are for an energy-optimised automotive pack, while a buffer discharged for a fraction of a minute is a different product optimised for the opposite thing. **That defence has since been tested against a measurement of the right product, and it does not survive.** A 24S nickel-cobalt-manganese pack was designed, built, bench-tested across discharge rates from 0.2 C to 10.68 C and then flown in an electric VTOL aircraft through six take-off, hover and landing cycles [47]. Its measured pack-level specific power is **724 W kg⁻¹** continuous for the unit pack and **892 W kg⁻¹** for the four-parallel flight system — inside the range quoted above, not above it. Its maximum tested rate, 10.68 C, corresponds to roughly **1.5 kW kg⁻¹**, and at that rate the pack reached 55.1 °C against a 60 °C limit: a thermal margin of 4.9 °C. That is the ceiling of a pack built for exactly this duty, instrumented, and flown.

A third source points the same way from the other direction. A NASA-funded design study of electrically propelled aircraft adopts a battery specific power of 4 kW kg⁻¹ and states plainly that this is "about **twice that of existing batteries**" [48] — that is, it treats a figure below the one assumed here as a future technology level rather than a present one.

Source	Specific power	Kind
Measured eVTOL pack, continuous [47]	0.72–0.89 kW kg ⁻¹	flight-tested measurement
Same pack at its maximum tested rate [47]	≈ 1.5 kW kg ⁻¹	measurement, 4.9 °C thermal margin
Li-ion for power applications [21]	0.7–1.3 kW kg ⁻¹	literature range
Assumed future level in a design study [48]	4 kW kg ⁻¹	stated as ≈ 2× what exists
This study, implicitly — hover	5.63 kW kg⁻¹	assumption
This study, implicitly — take-off	6.48 kW kg⁻¹	assumption

Sized at the measured thermal ceiling of 1.5 kW kg⁻¹ the buffer becomes **6.8 kg** rather than 1.8 kg — 5.0 kg heavier, against 2.2 kg of unallocated mass — and 7.8 kg if it must also lift the aircraft off the ground. Sized at the measured continuous figure it becomes 11.4 kg, or 23 percent of take-off mass, which is most of the payload. **The light design's mass budget does not close at any measured specific power, and this is the single most exposed number in the paper.** It is not resolved by arguing that the buffer is a different product: the source above *is* that product, built and flown. The gap to be closed is a factor of **3.8 on the measured thermal ceiling and 6.3 on the measured continuous rate**. What would resolve it is a pack demonstrating that, at acceptable

temperature, or a heavier buffer carried at the cost of payload fraction. Section 4 states which.

Both figures in this paragraph are larger than the ones an earlier version reported, and the reason is bookkeeping rather than new evidence: the buffer power was previously taken as a difference between two shaft stations and is now taken at the bus, where the buffer is.

The energy side is a different matter, and it is far more comfortable than the power side. The buffer does not discharge continuously for the whole vertical phase. It discharges through the take-off, which ends when the rotation begins and the wing starts carrying the aircraft; it then recharges from the engine over hours of cruise, and discharges again for the landing. At the achievable thrust-to-weight ratio of 1.132 the vertical acceleration is 0.132 g, so the 5 m s⁻¹ climb from which Section 3.15 enters the rotation is reached in 3.9 s and 9.6 m — and in 7.7 s and 19.3 m if full rotation authority is held in reserve. A take-off segment of ten to twenty seconds at full draw therefore remains generous, though less so than at the 1.2 an earlier version assumed. Against that:

Take-off segment at full draw	Energy used	of a 180 Wh kg ⁻¹ buffer	of an 80 Wh kg ⁻¹ buffer
10 s	32 Wh	10 %	23 %
20 s	65 Wh	20 %	45 %
30 s	97 Wh	30 %	68 %

These are the take-off draw of 11.66 kW at the bus. An earlier version of this table read 23, 46 and 69 Wh, which is 8.3 kW — the rotor-shaft-minus-engine-shaft difference retired at the head of this section. The correction survived into the power figures and not into the energy ones.

Even at the low energy density that a high-power chemistry would carry, a twenty-second take-off uses under half the buffer and a thirty-second one just over two thirds. **The binding constraint is power, not energy, and it stays that way under any plausible take-off duration** — though the margin on the energy side is thinner than the previous arithmetic suggested, and a buffer at 80 Wh kg⁻¹ would not tolerate a take-off much longer than the ones tabulated.

That distinction matters, because it says what would *not* rescue the specific-power figure. A shortfall in power is not a transient to be ridden out. Hover power goes as thrust to the three-halves, so thrust goes as power to the two-thirds, and a buffer delivering less simply buys less thrust:

Every row below is taken through the same chain: the buffer and the generator both deliver to the bus, and the bus is converted to propeller-shaft power by the power electronics and the electric machine before momentum theory is applied.

Buffer specific power	At the bus	At the propeller shaft	Resulting T/W	Leaves the ground
0.735 kW kg ⁻¹	3.66 kW	3.20 kW	0.44	no
1.30 kW kg ⁻¹	4.68 kW	4.09 kW	0.52	no
2.50 kW kg ⁻¹	6.84 kW	5.98 kW	0.67	no
5.63 kW kg⁻¹	12.47 kW	10.90 kW	1.00	only just
6.48 kW kg ⁻¹	14.00 kW	12.24 kW	1.13	yes

Two earlier versions of this table were wrong in two different ways, and both are worth naming because the second was introduced by the correction of the first. The original read 0.61, 0.71, 0.90 and 1.20 in the T/W column — a fifth too high at every row, because the column was scaled from an assumed T/W of 1.2 at its foot while 10.9 kW is, by the sizing statement of Section 2.12, the power for thrust equal to weight exactly. Correcting that left a second fault in place: the first three rows added the buffer's power to the engine's *shaft* rating and then read the sum as shaft power, which is the same mixing of stations this section has just spent two paragraphs removing from the headline figure. The rows above are on one formula throughout. The consequence of the pair of corrections is that the bottom of the table is worse than it looked: the specific power demanded rises, and what it buys is a hover rather than a take-off. Leaving the ground needs the tip pairs and 6.48 kW kg⁻¹.

At the specific powers this study has a citation for, the aircraft does not lift off, so there is no brief overshoot to tolerate and no manoeuvre that shortens the exposure. The only remedy is a heavier buffer, which is why this appears as a mass-budget exposure rather than a flight-dynamics one. The buffer masses used here are a cell-selection requirement rather than a free parameter, and it is the specific power, not the energy, that has to be found.

Group	Build-up	Assumed in 6.2
Structure	23.8 %	30 %
Propulsion chain	15.2 %	16 %
Battery buffer	3.6 %	4 %
Systems and contingency	11.0 %	8 %
Fuel	16.0 %	16 %
Payload, as residual	30.4 %	26 %

Where the mass may sit is also constrained, and the constraint was not previously stated. The budget above says how much each item weighs and not where it sits. Section 3.17 adopts a first-order packaging rule in the absence of an internal layout — masses distributed in proportion to internal volume — and shows that the resulting centre of gravity has to lie in a band whose aft limit is firm and whose forward limit is not: aft of roughly 85 percent of root chord the static margin falls below the usual tailless band, while the forward limit depends on how much camber moment the sections can supply, which this paper does not fix. The sweep carries the neutral point to 0.859 m from the root

leading edge, well aft of the root chord's midpoint, which is what makes such an aft centre of gravity admissible at all. The internal volume's own centroid is at 78.3 percent, so an arrangement that simply follows the available volume lands just forward of the window; fuel, payload and engine therefore cannot be placed for convenience, and the budget above should be read as constrained in position as well as in magnitude.

The build-up closes with 2.2 kg in hand — **conditionally, and the conditions are the result**. It closes if the average structural areal density is no more than 1.78 kg m^{-2} , and if everything still outside the model together stays under that same 2.2 kg. Neither is demonstrated here; the 1.5 kg m^{-2} used is an aggressive target for a composite airframe of this class rather than a measured property of one that has flown. Why a build-up coming in lighter than its own target should be read as a warning rather than a confirmation is set out in Section 4.4.

The heavy design is not closed by this exercise, and no claim is made that it closes. Shell mass scales as areal density times wetted area, so as the square of linear scale, while take-off mass scales as the cube; holding areal density constant would make the shell fraction fall as the inverse of scale, which is plainly wrong, since skins on larger aircraft are not thinner. Holding the fraction constant instead requires areal density to grow linearly with scale. The truth lies between, and the exponent has not been measured. Sweeping it puts the 1000 kg design's 260 kg payload at break-even at an exponent of 0.467, an areal density of 2.64 kg m^{-2} , closing below and failing above. No attempt is made here to argue for a value on either side of that threshold, because any such argument would be a structural model standing in for a measurement. What the sweep establishes is the statement itself: **the component build-up does not demonstrate closure of the heavy design**. That, and not any of the light-design assumptions, is the largest open question in the mass budget of this study, and it qualifies the scale-invariance of Section 3.9 — which holds for the analytical sizing fractions and has not been shown to hold for the structure that must realise them.

Supplementary S3 — Control axes in full

Supplementary material to "The Architectural Cost of Hybrid VTOL: meryemAircraft, a Propeller-Driven Tail-Sitting Blended-Wing-Body Without a Dedicated Lift System".

This is Section 2.10 of the paper, reproduced in full so that every control-axis number quoted in the main text can be traced to the calculation that produced it. The scripts that produce these numbers, and the record of the corrections made to them during the study, are in the repository the paper cites.

S3.1 Control without control surfaces

The nose pair produces thrust. The four tip pairs produce moments. They are not lift rotors and they are not sized to hover the aircraft; in the vertical phase they carry under fifteen percent of the total power. For the light reference design each tip pair is 0.20 m in

diameter and produces 16.2 N during the transition manoeuvre, drawing 335 W, for a total of 1.34 kW across the four.

The principle is not new, and the paper is better for saying so. A survey of tail-sitter development identifies it directly: "one way to generate a larger pitching moment to assist with the transition from vertical to horizontal flight is to add propellers away from the axis of rotation of the tail-sitter and apply a differential thrust to these propellers", and traces the idea to work at NASA Ames around the turn of the century [30]. What is particular here is not the mechanism but the arm and the accounting: the frames reach three tip chords rather than a fraction of one, and they are structure the aircraft needs anyway, so the arm is charged to the landing gear rather than to the control system.

The tip pairs sit at the ends of rigid frames that extend from each wing tip perpendicular to the planform, above and below, by three hundred percent of the local tip chord — 0.71 m in each direction, giving a vertical separation of 1.42 m between the upper and lower pairs. Figure 7 gives the placement and the resulting moment arms. The frames are long on purpose. The control moment is $M = 2 T L$, so lengthening the arm buys the same moment with less thrust; and because propeller power goes as thrust to the three-halves power, tripling the arm reduces the power required for a given moment to roughly one fifth. The frames are structure that is already needed for another reason, as Section 2.11 explains, so the arm is nearly free.

Pitch and yaw are produced by the same four actuators, but not with the same arm, and the difference has not previously been stated. Differential thrust between the upper and lower pairs produces a moment about the spanwise axis through the frame length, 0.71 m; differential thrust between the left and right pairs produces a moment about the remaining axis through the **semi-span, 1.726 m**. The yaw arm is therefore 2.43 times the pitch arm, and since the thrust available is the same, so is the moment: 55.9 N·m against 23.0 N·m at the quoted tip thrust, or 42.8 against 17.6 on the conservative thrust of Section 3.17. Yaw is the strongest axis this arrangement has, and it is strongest for a geometric reason rather than a designed one. In hover these are the two axes the aircraft must control against disturbance; in cruise, with the airframe rotated through ninety degrees, the same four actuators address the same two axes with their roles exchanged. No actuator changes its function, its orientation, or its mounting.

Roll is different, and this is the one place where propellers alone are not sufficient. The result is elementary but decisive. Every propeller on this aircraft has its thrust vector parallel to the body's longitudinal axis, so each thrust is a force $\mathbf{F} = (F_x, 0, 0)$ applied at a station $\mathbf{r} = (x, y, z)$. The moment about the longitudinal axis is

$$M_x = y F_z - z F_y = 0$$

identically, for every propeller, at every thrust setting, regardless of where it is mounted. No arrangement of parallel thrust vectors, and no number of them, can produce a rolling moment. The only roll moment available from the propulsion system is the residual reaction torque, which the counter-rotating arrangement of Section 2.9 has deliberately reduced to nearly zero. The two design decisions oppose one another, and the opposition is real rather than apparent.

This is the same limitation that constrains every tail-sitter, and it is worth being precise about how others resolve it, because the resolution here is not a variation on theirs but a consequence of a different decision made earlier.

Contemporary tail-sitters resolve it in one of two ways. Some place elevons in the propeller wash, which is a control surface by any definition. Others — including a carbon-fibre quadrotor tail-sitter that achieves full attitude control in every flight mode with no control surfaces at all — carry four separate single-rotation propellers and take their rolling moment from the *differential reaction torque* between them. That is the same mechanism a multirotor uses for yaw, appearing as roll once the airframe is rotated into wing-borne flight.

The second solution is unavailable here, and unavailable by construction. The coaxial counter-rotating arrangement of Section 2.9 exists precisely to cancel reaction torque, and it cancels the roll actuator along with it. The two decisions are not merely in tension, as noted above; they are mutually exclusive. A tail-sitter cannot both null its reaction torque and use that torque to roll.

This is the fork at which the present configuration departs from its nearest relatives, and it is why the strip is not an accessory. It is the element that makes the combination possible. The resolution adopted here is different in kind. A single strip on the lower surface, inclined at forty-five degrees, extends by a commanded amount rather than snapping between two positions. The extension is the control variable, so the rolling moment is modulated by how far the strip stands proud of the surface rather than by how long it is held out. For the light reference design it runs one hundred and twenty percent of the root chord in length, reaching outboard to sixty-seven percent of the semi-span, and stands 2 cm high at its inboard end and 6 cm at its outboard end.

Its authority comes from its length, not its height. The moment scales with the moment arm, whereas the benefit of additional height saturates: at constant length, doubling the height from 5 cm to 10 cm roughly doubles the roll rate, while extending the length from sixty to one hundred and twenty percent of root chord raises it almost fourfold. This comparison is robust to how the strip's force is modelled, because the length enters both the affected area and the moment arm while the height enters only the first.

Because extension is the control variable, the strip has a threshold, and the threshold has been measured. Tests of rearwardly located spoilers on two models established that "projections of less than $0.01c$ produce negligible changes in lift", and warned that a device with that property is poor as a control because "a small stick movement produces no change in trim, whereas a larger movement of the stick may produce large changes" [19]. The strip here is tapered, so the threshold is not crossed everywhere at once: height grows outboard while chord shrinks, and the outboard end reaches one percent of local chord at seven percent of full extension while the root does not reach it until forty-eight percent. Evaluated on the actual chord distribution, the strip is wholly inert below seven percent of commanded travel and wholly active above fifty.

Commanded extension	First active station	Active length	Effective arm	Rolling moment, fraction of full
0.05	none	0 %	—	0.000
0.10	0.945 m	19 %	1.057 m	0.041
0.15	0.668 m	43 %	0.930 m	0.113
0.20	0.481 m	59 %	0.850 m	0.177
0.30	0.239 m	80 %	0.757 m	0.293
0.40	0.090 m	92 %	0.707 m	0.399
0.50	root	100 %	0.679 m	0.500

The last column is the one that matters, and it is more benign than the threshold suggests. The part of the strip that survives the threshold is the part with the longest arm — the effective arm rises from 0.679 m at full extension to 1.057 m at one tenth of it — so most of the lost area is bought back by the lengthened arm. Above a quarter of travel the response is linear to within five percent; below fifteen percent more than a quarter of the commanded moment is missing; below seven percent there is none. **Continuous extension therefore does not give continuous authority from zero**, and a control design would have to carry that dead band explicitly. It bounds small corrections rather than large ones, which is the opposite of the usual complaint about a threshold device, and it is a consequence of the taper rather than of the concept: a strip of constant height fraction would cross the threshold everywhere at once.

What the roll axis costs, and what it is opposed by, are computed here. The roll axis had not been examined with the care given to pitch, and doing so separates a part that can be computed for this geometry from a part that cannot. Distributing the component masses of Section 3.11 by the same volume-weighted rule used for pitch gives a roll inertia of **25.0 kg·m²** — two and a half times the pitch inertia of 9.81 kg·m², because the mass is spread along the span rather than along the chord. Roll damping was then computed for this planform rather than taken from the literature: imposing the helix-angle twist $\theta(y) = -\arctan(p y / V)$ on a vortex-lattice solution of the actual geometry gives a damping coefficient of magnitude **|C_{l_p}| = 0.358**, linear in roll rate to 0.4 rad s⁻¹ and converged to within 1.3 percent over a threefold resolution refinement. At the cruise condition this is a damping slope of 77.6 N·m per rad s⁻¹ and a roll time constant of **0.32 s**, so the roll response is damping-dominated within a third of a second and the steady rate, not the initial acceleration, is what a control moment buys.

The requirement follows, and it is a requirement rather than a demonstration. A steady roll rate of twenty degrees per second at cruise needs **27.1 N·m**, and twenty-five degrees per second needs 33.9 N·m; time to a thirty-degree bank is then about 1.2 s. Whether the strip supplies that depends on which mechanism it works by, and the two candidates do not agree:

Mechanism	Rolling moment	Steady roll rate
The strip's own force, as a swept fence ($C_N = 1.3$, an upper bound)	11.3 N·m	8.4° s^{-1}
A change in the half-wing's circulation, $\Delta C_L = 0.10$	22.4 N·m	$16.6^\circ \text{ s}^{-1}$
the same, $\Delta C_L = 0.15$	33.7 N·m	$24.8^\circ \text{ s}^{-1}$
the same, $\Delta C_L = 0.20$	44.9 N·m	$33.1^\circ \text{ s}^{-1}$

The strip's own force cannot produce the twenty to twenty-five degrees per second this configuration needs — it falls short by a factor of about three, and an earlier version of this paper quoted 46 N·m without saying where it came from. The moment must therefore come from the second mechanism: the strip changes the circulation of the half-wing it sits on, which is how a Gurney flap or a low fence works, and the affected area is the wing's, not the strip's. Twenty degrees per second then asks for $\Delta C_L \approx 0.12$ over the strip's span.

That figure can be compared against a measurement, and the comparison is favourable. Traub [18] tested an aspect-ratio-three wing in a low-speed tunnel with a Gurney flap two percent of chord in height, in two spanwise arrangements: full span, and **the inboard two-thirds only**. The second is the same spanwise fraction the strip occupies here. From the measured lift-curve slopes and zero-lift angles of that inboard case, the increment at lift coefficients spanning the present cruise condition is:

Clean C_L	With inboard flap	ΔC_L
0.30	0.442	0.142
0.45	0.608	0.158
0.60	0.773	0.173

The requirement of 0.12 sits below the measured range rather than above it. **Two differences keep this short of a demonstration, and both cut the same way.** Traub's flap is at the trailing edge and perpendicular to the surface; the strip here is a swept fence on the lower surface, well forward of the trailing edge, and the mechanism is not identical. And his wing is unswept and rectangular where this one is swept and tapered. What the measurement establishes is that a device of this class, at this height fraction, over this spanwise extent, produces lift increments of the magnitude required — not that this strip produces one.

The comparison also exposes a design fault the paper had not noticed. Traub's flap is two percent of chord everywhere. The strip specified in Section 2.10 grows linearly from 2 cm to 6 cm while the chord it stands on shrinks:

Station along the strip	Local chord	Strip height	h/c
root	0.969 m	0.020 m	2.1 %
mid	0.682 m	0.040 m	5.9 %
outboard end	0.436 m	0.060 m	13.7 %

There is a measured threshold for this. A NASA study of lift-enhancing tabs — devices placed on the **pressure side** of a wing and **near rather than at** the trailing edge, which is the closest published geometry to the strip described here — reports that "for flap heights less than about 1.5 % c, the maximum L/D can also increase" while "flap heights greater than 1.5 % c cause a decrease in the maximum L/D" [32]. Every station on this strip is above that threshold and the outboard end is nine times it.

A law that held h/c constant would be a different trade, and the paper had not seen that there was one to make. Holding h/c at two percent everywhere gives a strip of 0.0165 m² instead of 0.0466 m² — sixty-four percent less frontal area, and the same reduction in deployed drag, from 16.7 N to 5.9 N at cruise. Because such a strip's area distribution follows the chord, its centroid sits at 0.507 m, exactly where the half-wing's own area centroid sits, so the rolling moment under the circulation mechanism is unchanged. What it costs is hover: the part of the strip lying inside the slipstream shrinks from 0.0182 m² to 0.0101 m², a forty-five percent reduction in whatever authority comes from the strip's *own* force at zero airspeed. Which way that trade should be settled depends on which mechanism dominates in hover, and this paper has not settled that. One thing that does *not* settle it is the L/D threshold above, because that threshold is about a device left deployed: this one is commanded on only while a roll is being flown, so its cruise-drag penalty is intermittent by construction and paying it is not obviously wrong. What can be said is that **the present height law was not derived from anything, and the alternatives are now visible.**

Deploying it on one side yaws the aircraft, and the paper had not said so. The strip raises lift on the half-wing that carries it and also raises drag there; the aircraft therefore rolls away from the strip and yaws towards it, which is adverse yaw in the classical sense. The magnitude follows from the same drag estimate: 16.7 N at the present height law, acting at 0.679 m, is **11.3 N·m** of yawing moment, falling to 2.9 N·m for a strip held at two percent of local chord. Against the 42.8 to 55.9 N·m of yaw authority computed above, the coupling costs between five and twenty-six percent of the yaw axis while a roll is being commanded. **It is covered, and it is covered by the axis that happens to be the strongest** — but it is a coupling, it had not been stated, and a control design would have to allocate for it.

It loads the pitch axis too, and that one is not comfortable. A wind-tunnel study of tapered-height Gurney flaps on a 60° delta wing — tested at heights of two and five percent of root chord, which is the same height law and the same range as the strip specified here — reports that "the flap significantly increases nose down pitching moment" [33]. The mechanism is the same on this aircraft: the lift the strip adds acts aft of the centre of gravity, so it pitches the nose down. How far aft depends on where along the chord the increment appears, and that is the part this paper cannot compute, since the

strip is neither at the trailing edge nor straight. Bracketing it by placing the increment at mid-chord, three-quarter chord and the trailing edge of the affected region:

Increment acts at	Moment about CG	ΔC_m
mid-chord	3.7 N·m	0.005
$\frac{3}{4}$ chord	13.2 N·m	0.019
trailing edge (upper bound)	22.7 N·m	0.032

The tightest pitching-moment budget in the whole of Section 3.17 is **0.050**, at the end of the rotation. At the upper end of this bracket, commanding a roll during that phase would consume two thirds of it. **The conclusion is a scheduling requirement rather than a redesign: the roll strip should not be commanded during the end of the rotation**, which is the one phase where the pitch axis has least to spare. The paper had not previously identified any reason to restrict when the strip may be used.

The coupling is not a novelty of this configuration, and the older literature states it more bluntly than the estimate above does. The same NACA survey reports that where spoilers were used for lateral control on tailless aircraft, "if only upgoing spoiler projections are used, the pitching moments developed are **prohibitive**", and that "a spoiler arrangement employing equal up and down projections would improve this condition" though the data then available were insufficient to settle it [19]. The strip specified here projects from one surface only, which is the arrangement that warning is about. Two things follow. The bracket computed above is a lower bound in kind as well as in magnitude, since it counts the lift increment and not the attendant pressure redistribution ahead of the device. And the remedy the survey names — projection from both surfaces, so that the pitching contributions oppose while the rolling contributions add — is available to this configuration in principle and has not been examined here. **The scheduling restriction above is the conservative resolution; a two-sided strip would be the structural one, and choosing between them needs the wind-tunnel measurement this section has already asked for.**

The same device has a third use, and it is the one that costs nothing. Deployed on both halves at once the strip is not a roll control but a speed brake, and in that mode the pitching-moment objection does not arise: a wind-tunnel investigation of spoiler-type ailerons used as speed brakes and glide-path controls found that they "had only a small effect on the wing pitching moments", and — the part that matters for a configuration with one moving surface — that "the rolling effectiveness of the ailerons will not be impaired by such use" [49]. A tail-sitter lands vertically and does not need a glide-path control for that, but it descends to the transition point like any other aircraft, and the strip gives it a descent-rate control that no other part of this configuration provides. The transfer is directional only: those tests are on conventional wings with slotted flaps at higher Mach number, and the device projects from the upper surface rather than, as here, the lower.

Roll authority is therefore sized, supported by a measurement on a comparable device, and still not closed. The quantity a future measurement must return is ΔC_L for this strip on this planform, not a moment.

Hover is the harder case, and for a reason that is structural rather than numerical. At zero airspeed only the inboard part of the strip is loaded, by the slipstream, and the same circulation model over the slipstream-washed area gives 6.0 to 12.0 N·m for ΔC_L between 0.10 and 0.20 — enough for a thirty-degree bank in 1.5 to 2.1 s. But at zero airspeed there is no aerodynamic damping at all: the roll axis is a double integrator, so the rate does not settle and the strip must be commanded off rather than left on. Roll control in hover is consequently a tighter problem than roll control in cruise, which is the reverse of the usual situation and is a consequence of this configuration rather than of its numbers.

The worst case, in which the extension is not modulated at all, is worth computing because it bounds the actuator requirement. Simulating the first-order roll dynamics above as though the strip snapped between fully out and fully retracted, with a deadband and a finite actuator delay, gives a bounded limit cycle: with a two-degree deadband and a fifty-millisecond deployment the bank angle oscillates by $\pm 0.2^\circ$, and at a hundred and fifty milliseconds by $\pm 9.4^\circ$. **A proportional extension does not produce that limit cycle at all**, so these figures are an upper bound on what the actuator must achieve rather than a description of how the aircraft is flown — and they say that even a device reduced to two positions would be usable provided it were fast. None of this is a closed-loop stability analysis, and none of it substitutes for one.

Two consequences of variable extension are worth stating, because both relieve constraints identified above. The couplings scale with the extension: adverse yaw and the nose-down pitching moment are proportional to the lift and drag increments the strip produces, so a roll commanded at a fraction of full extension carries that fraction of both. The scheduling restriction derived above — that the strip should not be commanded at the end of the rotation — is therefore a restriction on *full* extension, and small corrections remain available throughout. And the height law is limited by those couplings rather than by mass: the strip's own structure is a small item, so how far it may extend is set by how much yaw and pitch disturbance the tip propellers can absorb, which is the calculation given above rather than anything in Section 3.11.

Yaw was examined last, and it separates cleanly into an easy half and an open half. The easy half is authority. The yaw inertia computed from the same mass distribution is 33.7 kg·m² — close, as it must be for a nearly planar aircraft, to the sum of the roll and pitch inertias — so the available yaw moment turns the aircraft's own inertia at 73 to 95 degrees per second squared, reaching fifteen degrees of heading in about six tenths of a second. Nothing in this axis is short of moment.

The open half is stability. A vortex-lattice solution of the planform at sideslip returns $C_{n_\beta} = 0$: the wing supplies no directional stability whatever, which is not a defect of the solution but the expected result for a planar surface with nothing to generate side force. Two measurement campaigns say the same thing about real aircraft rather than about a panel model. A NASA series of four 60°-swept flying wings tested from -8° to 48° of incidence found that, without vertical tails, "each of these wings possessed unstable or essentially neutral values of directional stability for most of the angles of attack tested" [20]; and the NACA survey of tailless practice records that "the directional stability at low

angles of attack for the wing alone has generally been found to be inadequate" [19]. **The present configuration is not unusual in lacking weathercock stability; it is normal, and every tailless aircraft that flies has had to buy it somewhere.** Sweep gives this configuration its roll-due-to-sideslip — $C_{l_\beta} = -0.045$ per radian, a healthy value — and gives it no weathercock stability at all. The profile-drag contribution to yaw damping is likewise negligible, $C_{n_r} = -0.0023$, a time constant of over a minute. That last number is less alarming than it looks, and the reason is conditional. Low rotational damping is inherent to the tailless class, and free-flight experience is that the small values "will not be excessively detrimental to the flying qualities **provided the directional stability of the airplane is adequate**" [19] — so the two open items in this axis are not independent, and the yaw-damping deficit is forgiven only if the weathercock requirement below is actually met. The same source adds that lateral-oscillation damping is most critical at high speed, because both C_{n_r} and the coupling between yawing and rolling diminish at low incidence. **For this aircraft that names cruise, not transition, as the critical case for Dutch roll.**

And zero is an optimistic starting point, not a neutral one. A vortex-lattice model has no volume, so the calculation above contains the planform and nothing else. The centre body of a blended wing is a body, it develops side force in sideslip, and that side force acts forward of the centre of gravity. The survey states the magnitude in the only terms that matter here: on tailless aircraft the destabilising effect of the fuselage and nacelles "is usually at least as great as the stabilising effects contributed by the wing alone" [19]. The wing alone here contributes zero, so the comparison gives no number — but the sign is unambiguous, and the true figure the fairing has to make up is **below** zero rather than at it. Nothing in this section quantifies that deficit, and the margin reported below should be read with it outstanding.

Directional stability must therefore come from the tip frames, which in cruise stand perpendicular to the wing plane above and below each tip and are the only vertical surfaces the aircraft has. Their mid-chord sits 0.879 m aft of the centre of gravity, which is a long arm for a surface that already exists. Taking a lift-curve slope of 4 per radian for a slender faired strut — an assumed value, not a measured one — the side area needed is 0.058 m² to reach $C_{n_\beta} = 0.03$ and 0.097 m² to reach 0.05, which spread over the 2.84 m of combined frame length is a fairing chord of 21 mm and 34 mm.

Those two targets were chosen by this study rather than taken from anywhere, and there is a published criterion that should have been used instead. The NACA survey of tailless practice states that tailless aircraft are **not exempt** from the conventional standard — directional stability "should be as great as required on conventional airplanes if the same requirements regarding satisfactory flying qualities are to be adhered to" — and puts that standard at C_{n_β} "usually greater than **0.001 per degree**", noting that free-flight tunnel models were flown successfully at one third of it but that "the best flying qualities of these models were obtained with values of C_{n_β} in excess of 0.001" [19]. In radian measure the recommendation is **0.0573** and the demonstrated floor is 0.0191. Both values this study picked lie below the recommendation.

Sized against the criterion rather than against a guess, the fairing chord required is **39 mm**, and the floor demonstrated in free flight is met at 13 mm. A faired strut of the 20 mm thickness assumed in Section 3.3 carries a chord of several times its thickness — typically 50 to 70 mm — so the established criterion is met by a fairing **smaller than the one the structure needs anyway**, with between a quarter and three quarters of that chord left over. The conclusion the paper drew from its own two numbers survives being held to a real standard, which is the only reason it is worth restating: directional stability on this configuration does not ask for a surface, it asks for a fairing on a frame that is already there.

Where to buy it is a question the tailless literature has already answered, and the answer is the arrangement this aircraft already has. The NACA survey states that "if the tailless airplane has a swept-back wing, the usual practice is to place the vertical tail surfaces at the tips rather than at the center section in order to take advantage of the longer moment arm" [19]. That is the tip-frame arrangement of Section 2.9, adopted here for the moment arm it gives the control propellers and for the landing structure it provides, and it turns out to be the placement a directional-stability surface wants for an independent reason.

The same source adds two things the estimate above does not contain, and they pull in opposite directions. The first is favourable: because a tip fin's drag acts at a moment arm of half the span, "the drag characteristics as well as the lift characteristics of the tip fins exert an influence on the directional stability", and fins working on the profile-drag principle were found more effective than those working on lift. The area estimated above was sized on lift alone, so it is an **over**-estimate of what is needed. The second is a requirement: the toe angle is not free. Fins of aspect ratio below about two must be toed *in*, and fins of moderate or high aspect ratio toed *out* — and the frames here, at 1.42 m long against a fairing chord of tens of millimetres, are firmly in the second class. Toe-out carries a hazard the paper had no way of knowing about: yawing far enough to stall the rear fin produces a large *destabilising* moment, where a toed-in fin stalling produces a stabilising one.

The magnitude, at least, is small. Wind-tunnel tests of a tail-less swept cargo configuration with vertical tails carried on tip pods found the optimum toe angle for maximum lift-to-drag ratio to be **about 1.5 degrees** for symmetric-section tails, with a cambered tail "relatively insensitive to toe-in" [34]. The same tests found the lift-to-drag ratio "about the same with all three vertical-tail designs, notwithstanding the 75 % larger area" of the largest — which is the same conclusion reached above from the required-area side, arrived at by measurement rather than by estimate. **The frame fairing therefore needs a toe angle of order a degree or two, and that angle needs a sign.**

An earlier version of this paper left the sign open on the ground that these frames lie beyond the aspect-ratio range of either source. That reasoning was wrong, because the rule is not a correlation but a statement about which force does the work. Low-aspect-ratio fins are toed *in* so that the stabilising moment is generated by the large induced drag such surfaces carry; high-aspect-ratio fins are toed *out* so that it is generated by "the outwardly directed lift" [19]. Raising the aspect ratio shifts the balance further from drag and

towards lift, so it strengthens the case for toe-out rather than carrying the rule outside its range. **The sign is toe-out.**

What is genuinely open is narrower and more awkward. A fairing of 39 mm chord at cruise speed sits at a Reynolds number near **80,000**, and the assumed lift-curve slope of 4 per radian is an assumption at any Reynolds number and an optimistic one at that. Symmetric sections at low Reynolds number are measured to behave badly in exactly the band a toe angle of one or two degrees occupies: of four symmetric sections tested at Princeton, all showed lift-curve nonlinearity about zero incidence, and in a more severe case the slope of the lift curve "actually changed sign over a 3 deg range" [36].

That is not one section behaving oddly; the same compilation states it as a property of the class. Introducing a section designed for horizontal tails, it gives the reason such a section is cambered at all: "past work on symmetrical airfoils has shown that a deadband often appears in the lift curve near zero degrees. This nonlinearity can lead to undesirable longitudinal handling characteristics. Interestingly, cambered airfoils do not appear to have a similar, intrinsic deadband region" [36]. Elsewhere it records a deadband at Reynolds numbers of 60 000 and 100 000 on a section where one was not expected, noting that "this type of behavior is usually only seen on symmetrical airfoils at low Re's" and that "at higher Re's the dead band is not present" [36]. The fairing sized here is a symmetric section at 80 000, which is inside that range, and it is asked to work at one to two degrees, which is inside that band.

The question is therefore not which way to toe the fairing but whether a surface of that chord, at that Reynolds number, develops the side force this section has credited it with at all. Two things follow. The assumed 4 per radian is an upper bound and not a conservative choice, because what the measurements remove is the linearity of the curve rather than only the magnitude of its slope; a directional-stability margin computed on a linear derivative through zero is computed on the one part of the curve the data say is not there. And the remedy the source itself names is available at no structural cost: camber the fairing outboard and keep the toe-out, so that the operating point sits on the linear part of a curve that has one. This paper does not size that fairing. Doing so would mean choosing a slope from a curve it has not measured, which is the error the preceding paragraph exists to record.

This is the measurement this configuration would buy first: side force and yawing moment on a faired tip frame of the geometry of Section 2.11, symmetric and cambered, through small sideslip, at a chord Reynolds number of 80 000. It is a small model in a small tunnel, and it would either confirm the only directional-stability surface this aircraft has or remove it.

That same study is the third independent report of the finding this subsection began with: a podded tailless wing was directionally *unstable*, and "the further addition of vertical tails restores directional stability to the configuration" [34].

That reframes the fairing, and the reframing is the substantive result of this subsection. Section 3.3 introduced the fairing as a drag measure and computed the frame drag penalty on that basis. It is also, and not incidentally, the aircraft's directional stability surface and its principal source of yaw damping. The two roles are served by the same

hardware — which is the same pattern the whole paper is about — but the paper had not noticed the second role, and the fairing's chord is consequently constrained from two directions rather than one. Neither the fin contribution nor the yaw damping it brings is computed here. **Yaw authority is sized; directional stability is a requirement placed on a component the design already carries.**

Two measured effects work against the strip in exactly the conditions this configuration needs it, and both were found by reading rather than by calculating.

Wind-tunnel tests of a Gurney flap under controlled inflow turbulence report that the device "became less effective after stall angle", and that at a turbulence intensity of 19 percent "the benefit to the aerodynamic performance was negligible", against gains of 2.7 to 14.4 percent in lift-to-drag ratio at 10.5 percent [35]. Both findings point at this aircraft. The transition passes through incidences at which the outboard wing is post-stall, which is where roll disturbances are largest and where the device is reported to weaken; and the strip's inboard portion is deliberately placed inside the nose propeller's slipstream, which is not a low-turbulence environment. **The slipstream placement that gives the strip authority at zero airspeed may also be the place its mechanism works least well.** Neither effect is quantified for this geometry — the cited tests are on a wind-turbine aerofoil with grid-generated freestream turbulence, not a propeller wake — but the direction is measured rather than supposed, and it is recorded in Section 4 as a risk to hover roll authority specifically.

On the post-stall half of that risk the literature is not unanimous, and reporting only the unfavourable half would misrepresent it. A delayed-detached-eddy simulation of a 21 %-thick section at twenty degrees of incidence — deep stall — with a Gurney flap of two percent chord, which is the height fraction at this strip's root station, returns a lift coefficient higher by ninety-four percent with the drag coefficient unchanged to within half a percent [37]. That is the opposite of a device weakening past the stall. The two results are not directly comparable: one is a measurement and the other a simulation whose plain-aerofoil baseline falls twenty-five percent below the experimental lift at the same condition, and they vary different things — one inflow turbulence, the other incidence. What can be said is that **the post-stall behaviour of this class of device is contested, while the turbulence sensitivity is measured and stands unopposed.** The risk carried into Section 4 is therefore the turbulence one primarily, with post-stall behaviour an open question rather than a known deficit.

A third result bears on the part of the claim that looked weakest. The same device was visualised in a water tunnel at a Reynolds number of 8 588 — four orders of magnitude below the wind-tunnel work — and behaved in qualitative agreement with it, the authors attributing this to the mechanism being an effective camber increase and therefore "an inviscid effect to the first order" [38]. **That is the most direct support available for the claim that the strip still works at zero airspeed,** where the only flow it sees is the slipstream and the Reynolds number is at its lowest. It supports the direction and not the magnitude: the study is flow visualisation and reports no forces. The same source also records the cost, measured rather than estimated: a strip of 1.25 percent chord leaves drag unchanged from the clean wing, while one of five percent brings a significant drag increase [38]. This strip runs from 2.1 to 13.7 percent.

The strip does one further thing that an ordinary aerodynamic surface cannot. Its inboard portion lies inside the slipstream of the nose propeller, where the dynamic pressure is set by the disc loading rather than by the airspeed:

$$q_{\text{slipstream}} = T / A$$

For the light reference design this is 433 N m^{-2} in hover — the dynamic pressure of a 26 m s^{-1} freestream — available at zero airspeed. The strip therefore produces a usable moment while the aircraft is standing still, which an aerodynamic surface outside the slipstream cannot. Its outboard portion lies beyond the slipstream, where it works against the freestream in cruise. The slipstream covers only twenty-seven to thirty-nine percent of the semi-span, so lengthening the strip to serve cruise does not compromise its hover function; one device serves two regimes. Figure 8 shows the strip against the slipstream boundary: the inboard 46 % of its length lies inside, the outboard 54 % outside.

Supplementary S4 — Rotational authority, trim, and the transition envelope

Supplementary material to "The Architectural Cost of Hybrid VTOL: meryemAircraft, a Propeller-Driven Tail-Sitting Blended-Wing-Body Without a Dedicated Lift System".

This is Section 3.17 of the paper, reproduced in full so that every figure in the trim chain and the rotational-authority budget can be traced to the calculation that produced it — including the inertia derivation, the rotation profiles, the centre-of-gravity window, the twist sweep, and the measured reflex-section evidence. The scripts are in the repository the paper cites.

S4.1 Whether there is enough authority to rotate

Section 3.15 drives the body angle kinematically. The aircraft does not rotate in that simulation; it is *assumed* to rotate, and the moment producing the rotation does not appear. Supplementary S5.6 records this. What follows does not remove that limitation — a full six-degree-of-freedom treatment would need pitching-moment coefficients through ninety degrees of incidence, and no such data exists for this planform — but it closes the part of the question that can be closed without them.

The test is a lower bound. If the tip propellers cannot rotate the aircraft's inertia, they certainly cannot rotate it against aerodynamic moment as well. If they can, the aerodynamic margin remains unknown and is reported as unknown.

Inertia, and where the mass may sit. The component build-up of Section 3.11 supplies the masses; the planform of Section 2.8 supplies where they can go. The shell and internal structure are distributed over the planform, the tip frames along their own length, the tip motors and propellers at the ends of those frames. The coaxial pair, its hub and the

electric machine that drives it are fixed at the nose. Everything else — engine, generator, power electronics, fuel, battery, avionics, payload — is distributed **in proportion to the internal volume available to hold it**, which is the constraint a real internal arrangement faces and is not a free choice.

That distinction matters more than it appears. An earlier version of this calculation placed those items along the root chord by hand and obtained a centre of gravity at 57 percent of root chord. That was wrong, and the reason is the sweep: the outboard sections lie well aft of the root trailing edge, so the internal volume's own centroid sits at 78.3 percent of root chord and the structural centroid at 99 percent. Placed where the volume actually is, the centre of gravity falls at **80 percent of root chord** for the light design and 82 percent for the heavy one, and the moment of inertia about the spanwise axis — the axis the transition rotates about — is **9.81 kg·m²** and **2 503 kg·m²**. The hand-placed figures were 40 and 30 percent lower, and the margins below are correspondingly tighter than that earlier version reported.

The rotation profile matters, and Section 3.15's cannot be produced. That simulation ramps the body angle linearly, which requires zero torque throughout and infinite torque at each end. Two profiles a finite moment can produce bracket the choice: a bang-bang profile, accelerating for the first half and decelerating for the second, needs a peak angular acceleration of $4\Delta\theta/t_r^2$, and is the cheapest rest-to-rest profile there is; a smooth profile bringing angular velocity and acceleration to zero at both ends needs $6\Delta\theta/t_r^2$. A feasibility test — whether the manoeuvre is possible at all — must use the cheaper of the two, so the bang-bang value is taken as the requirement and the smooth value is reported alongside it as what a gentler command would cost.

Authority. The frames place the upper and lower pairs 0.71 m from the planform in the light design, and differential thrust between them acts about the spanwise axis, as Section 2.9 sets out. Propeller thrust on this aircraft cannot reverse, so the largest differential available is the upper pairs at full thrust against the lower pairs at zero, which is the $M = 2TL$ of Section 2.9 and not four times the single-pair moment. The available moments are therefore 23.0 N·m for the light design at its quoted 16.2 N per pair, and 952 N·m for the heavy design, whose transition thrust is not quoted in Section 3.8 and is computed here from its twelve percent power share as 200 N per pair.

	Required, bang-bang	Required, smooth	Available	Margin, bang-bang	Margin, smooth
Light, $t_r = 2$ s	15.4 N·m	23.1 N·m	23.0 N·m	1.49 ×	0.99 ×
Heavy, $t_r = 5.1$ s	605 N·m	907 N·m	952 N·m	1.57 ×	1.05 ×

The reference rotation times are actuator-limited lower bounds, not comfortable choices. Both designs close on the cheapest rest-to-rest profile with a margin near 1.5, and both sit essentially at unity on a smooth one: 2.01 s and 4.98 s are the shortest smoothly-commanded rotations the tip propellers can force, against reference times of 2 and 5.1 s. Read correctly, that is not a margin of five percent — it is the statement that the

reference rotation *is* the minimum smooth rotation the moment authority allows, to within the precision of the thrust estimate, and that anything faster is available only by commanding a profile with discontinuous angular acceleration. The two designs are at the same point on the same constraint, which is the tip-propeller moment; the transition times follow from it rather than being chosen. A design iteration wanting genuine comfort on a smooth command would lengthen both rotations by about a quarter, which Section 3.15 shows costs nothing in altitude — and that, rather than the quoted times, is what a design study should carry forward.

The field has a name for what this section computes, and a settled opinion about where in it to fly. The feasible set of transition states is called a *transition corridor*, and it is used to turn trajectory generation over a complex aircraft model into a constrained motion planning problem [31]. The envelope of Section 3.15 and the moment limits here are a corridor of that kind, computed rather than borrowed. On where to fly inside it, the same source is explicit that existing corridor-based studies "only try to plan the flight trajectory in the middle of the corridor, considering that the corridor bounds might be sensitive to aerodynamic uncertainties and disturbance". **The reference rotation times of this paper are on the bound, not in the middle** — which is exactly the practice that source warns against, and an independent reason to read the recommendation of the preceding paragraph as the design statement and the quoted 2 s and 5.1 s as the limit they approach.

This calculation set the heavy design's rotation time. An earlier version of this study used four seconds. Against the inertia of this section that gives margins of 0.97 on the cheapest profile and 0.65 on a smooth one — that is, infeasible on both, since the shortest rotations the moment authority allows are 4.06 s and 4.98 s. Four seconds was not a margin and, with the corrected inertia, not even a boundary. Lengthening the rotation to 5.1 s brings the heavy margins to 1.57 and 1.05, matching the light design's 1.49 and 0.99 at its quoted two seconds, and costs nothing: Table 4 shows the tip-propeller power falling from thirteen percent of hover power to six, and the altitude-loss result of Section 3.15 is unchanged, remaining zero at every profile tested when the rotation is entered in a climb. The light design's own two seconds is on the same boundary — its smooth minimum is 2.01 s — so neither reference design has margin to spare on a smooth command, and both should be read as sized by this constraint.

This is also, in moment terms, what Table 4 of Section 3.9 already said in units of power: that four seconds consumed nearly the whole tip-propeller allocation. The two statements agree, and the present calculation adds the rotation profile, which Table 4 did not distinguish.

The light figure depends on a thrust the paper quotes without a basis. At 335 W and 0.20 m diameter, 16.2 N implies a figure of merit of 0.702 with no coaxial interference loss, where the hover figure of merit used elsewhere is 0.599. The disc area used is that of one rotor rather than two, which is the ordinary treatment of a coaxial pair — the lower rotor works in the upper's wake, so the pair is charged a single disc and an interference penalty rather than twice the area.

Recomputing at 0.599 with a fifteen percent coaxial loss gives 12.4 N and an available moment of 17.6 N·m, which against the bang-bang requirement of 15.4 N·m is a margin of 1.14 and against the smooth requirement of 23.1 N·m is 0.76.

An earlier version of this section reported 1.59 here and called it comfortable; that figure does not follow from 17.6 against 15.4, and the correct one is not comfortable. On the conservative thrust basis the light design closes only on the cheapest rotation profile and only just, and the smooth profile does not close at all at two seconds. The heavy figure was computed on that conservative basis to begin with, which is why the two designs are not directly comparable on this row. Section 4 carries the consequence: the reference rotation times are actuator-limited lower bounds under either thrust basis, and under the conservative one the light design has no margin to give.

The margin narrows with size, and the narrowing is measured rather than derived. An earlier version of this section derived a scaling law by assuming geometric similarity. The two reference designs are not geometrically similar: span grows by a factor 3.345 while mass grows by twenty, and 3.345^3 is 37.4, not 20 — the wing loading rises from 25.3 to 45.0 kg m⁻² precisely because they are not. The ratios are therefore read from the two computed designs instead. Inertia grows by 255.1 while available moment grows by only 41.4, so the margin is preserved when required moment grows by the same 41.4 — which fixes the rotation time at 4.96 s, and is where the 5.1 s of Section 3.8 comes from. Section 3.15 already concluded that the larger aircraft must rotate more slowly; the quantitative form of that statement is that the rotation time must grow as the square root of the ratio of inertia growth to moment growth, and that setting it any faster spends control margin to buy nothing, since a slower rotation loses no more altitude. **What this does not establish.** The centre of pressure moves as the aircraft rotates, and the pitching moment that produces is not computed here. The margins above are inertial: they say the propellers can turn the aircraft's own inertia and say nothing about turning it against aerodynamic moment. This is a necessary condition, not a sufficient one.

How much is left over can be resolved along the trajectory, and doing so corrects the question. The moment remaining after the inertia is turned — 11.9 N·m for the light design and 489 N·m for the heavy one — divided by $q S \bar{c}$ gives the pitching-moment coefficient that would consume it. Evaluating that along the trajectory of Section 3.15 rather than at a single representative speed shows something the single-speed form obscured: **the aircraft does not reach ninety degrees of incidence.** The body rotates through ninety degrees, but the relative wind rotates with it, because the aircraft is accelerating and climbing at the same time. Peak incidence is 17.5° for the light design entered in a 5 m s⁻¹ climb, and 21.6° entered from rest.

	Entry	Peak incidence	Airspeed there	C _m budget there	Tightest budget, and where
Light	5 m s ⁻¹ climb	17.5°	7.3 m s ⁻¹	0.205	0.050, at rotation end, $\alpha = 4.9^\circ$, 14.7 m s ⁻¹
Light	from rest	21.6°	2.8 m s ⁻¹	1.381	0.051, at rotation end, $\alpha = 17.7^\circ$, 14.7 m s ⁻¹
Heavy	5 m s ⁻¹ climb	5.4°	35.6 m s ⁻¹	0.010	0.010, at rotation end
Heavy	from rest	20.5°	6.8 m s ⁻¹	0.287	0.011, at rotation end, $\alpha = 6.5^\circ$, 35.4 m s ⁻¹

These budgets are what remains of the tip-propeller moment after the inertia is turned, so they follow the corrected inertia of this section. An earlier version of this table used an inertia thirty-nine percent lower and reported budgets correspondingly larger — 0.322 at the light design's peak incidence rather than 0.205. The values above supersede it.

The constraint splits into two, and they are different problems. In the middle of the rotation the incidence is high — seventeen to twenty-two degrees — but the dynamic pressure is low, and the coefficient that would consume the margin is 0.205 entering in a climb and 1.38 entering from rest. An earlier version of this paper compared these against a generic post-stall range quoted for swept planforms; that comparison is dropped, because the trajectory-resolved budget above supersedes it and the generic range was not attributable to a source we had read. What the budgets say without it is enough: entering the rotation from a climb buys altitude at the cost of arriving at the high-incidence phase faster, and therefore with less moment to spare, and 0.205 is not a comfortable coefficient to have left. At the end of the rotation the incidence is small, five to six degrees, but the aircraft is fast, and the budget falls to 0.050 for the light design and 0.010 for the heavy one. That second condition is **not** a post-stall problem: it is the trim question of a tailless aircraft at its cruise incidence, and it is examined below rather than asserted — though examining it separates a part that is shown from a part that is not.

The centre of gravity is an assumption, and it is stated as one. No detailed internal layout exists for this aircraft, so none can be measured. In its place a first-order packaging rule is adopted: the non-structural masses — fuel, payload, engine, generator, buffer — are distributed in proportion to the internal volume available at each station, and the structural mass in proportion to the shell area. Applied to the planform of Section 2.8 this rule puts the centre of gravity at **0.778 m aft of the root leading edge**, which is 80.2 percent of root chord, or 21.9 percent of mean aerodynamic chord aft of the mean-chord leading edge. Every stability figure below follows from that rule and not from a layout; a different arrangement of the same masses gives a different centre of gravity, which is why the sensitivity to it is tabulated rather than assumed away. Burning the full fuel load moves the centre of gravity aft by 0.3 points of root chord, which is inside the tabulated window and is therefore not a separate constraint.

Static stability is shown. A vortex-lattice solution over the planform of Section 2.8 places the neutral point at 0.859 m from the root leading edge — 34.4 percent of mean

aerodynamic chord, an entirely conventional value — and the result is converged, moving by 0.26 percent over a threefold refinement. With the centre of gravity where the packaging rule puts it, the static margin is **+12.5 percent of mean aerodynamic chord**, in the middle of the usual tailless band of five to fifteen percent. The configuration is statically stable in pitch, and it owes that to the sweep, which carries the neutral point aft faster than it carries the volume. All chord-referenced quantities here use the true mean aerodynamic chord, 0.651 m; an earlier version of this section quoted the margin on the mean aerodynamic chord but the trim requirement on the mean geometric chord, 0.573 m, and the two are now on the same datum. Because that error was a mismatch of conventions rather than of arithmetic, the absence of a second one was checked rather than assumed: the two chains — centre of gravity to neutral point to static margin, and centre of gravity to aerodynamic moment to required trim coefficient — were tested for a common origin, a common sign convention, independence of dynamic pressure, and agreement between the coefficient route and a dimensional route that never forms a coefficient at all. All four hold; in particular $W(x_{np} - x_{cg})$ and $C_L \times (\text{static margin}) \times q S \bar{c}$ give the same 39.82 N·m. Re-deriving the margin directly from the solver's moment about the centre of gravity, rather than from the neutral point, gives 12.8 percent against the 12.5 quoted above; the 0.3-point spread is the curvature of the fitted lift-moment slope and is smaller than the spread across the centre-of-gravity window.

Trim was an open item, and it is now closed — by twist rather than by camber, and at a price. At the cruise lift coefficient of 0.45 the moment to be balanced about the centre of gravity has coefficient $C_L \times (\text{static margin})$: 0.056 at the centre of gravity the packaging rule gives, and less further aft.

CG, % root chord	x_{cg} , m	Static margin, % MAC	Moment to be balanced
78	0.757	+15.7	0.071
80.2 (packaging rule)	0.778	+12.5	0.056
83	0.805	+8.3	0.037
85	0.825	+5.3	0.024

Two published benchmarks place that window, and both place it favourably. A blended-wing UAV of this class reports its own static margin of 0.081 of the mean aerodynamic chord as "marginally outside the typical range for static longitudinal stability", which it gives as 0.1 to 0.3, and its C_{m_α} of -0.086 per radian against a typical range of -0.3 to -1.5 , concluding that stability augmentation "through either forward movement of the centre of gravity or additional reflex of the airfoils is advisable" [41]. The configuration here sits inside both ranges at the packaging centre of gravity: 0.125 of the mean aerodynamic chord, and $C_{m_\alpha} = -0.48$ per radian on the lift-curve slope of 3.86 per radian the same solution returns. The second of those inherits the magnitude caveat of Section 3.10 — but that caveat runs the safe way here, since a lift-curve slope under-stated makes C_{m_α} under-stated too, and the true value would lie further inside the range rather than outside it. **The aft limit of the window, at 5.3 percent, is the one that falls outside the published range**, which is a second reason beyond static margin alone to prefer the packaging value to the aft end of the window.

An earlier version of this paper supposed that reflexed sections would supply this, on the strength of a range its authors had not read. Reading the source settles it in the other direction. In the variable-density tunnel measurements of Jacobs, Ward and Pinkerton [17], the reflexed section NACA 2R212 — two percent camber, mean line shaped specifically to give a small positive moment — returns $C_{m0} = +0.004$, against -0.002 for the symmetric 0012 and -0.044 for the conventionally cambered 2412. That is one fourteenth of what the centre of gravity above demands, and one sixth of what even the aftmost entry in the table demands. The same report concludes that reflexed mean lines "may be of questionable value because of the adverse effect of this mean-line shape on the maximum lift coefficient." **Reflex of the magnitude that has actually been measured does not trim this aircraft.**

2R212 is not an isolated case, and the wider measured record is worse for reflex than that single number suggests. Three further families were tested in the same variable-density tunnel, with pitching moment measured rather than computed, and none of them returns a positive value:

Section	c_m at zero lift	Test
NACA 2R212	+0.004	Variable-density tunnel [17]
Boeing 106R	-0.001	Variable-density tunnel, $Re \approx 3.1 \times 10^6$ [45]
Navy 60R	≈ -0.001	as above [45]
Göttingen 398R	-0.007	as above [45]
NACA M6	-0.001	as above [45]
NACA 4409R	-0.025	Variable-density tunnel [46]
NACA 4412R	-0.030	as above [46]
NACA 4415R	-0.031	as above [46]
NACA 4418R	-0.030	as above [46]

The three 1930s reflexed sections of the second group were built by replacing the mean line aft of thirty percent chord with a curve chosen, from thin-aerofoil theory, to give **zero** moment about the quarter chord — and the measurements report their moments as "practically zero" over the useful incidence range [45]. The 4400R family was designed to a target of -0.03 and the report states that "the design pitching-moment coefficient was realized" [46]. **Reflex, as actually built and measured, is a device for removing negative pitching moment, not for producing positive pitching moment.** Across nine measured sections the single positive value is $+0.004$, and the requirement here is $+0.056$.

An earlier version of this paper wrote that no plausible amount of reflex would trim this aircraft, then withdrew that as too wide on the strength of a published optimisation which trims a blended-wing-body of this class by treating reflex as a design variable and closing the chain with only **2.44 degrees** of twist [41]. The withdrawal went too far in the other direction. That study publishes no moment coefficient for the sections it generates, so the inference that they supply much more than 0.004 was an inference and not a reading; the

measured record above runs against it. What the comparison does establish is narrower and still worth saying: that study's trim requirement is the lighter one — a static margin of 0.081 of the mean aerodynamic chord against 0.125 here — and reflex was not asked to act alone.

A flying tail-sitter shows where the positive moment actually comes from. The aerodynamic model identification of a VTOL tail-sitter describes its own trim state explicitly: the aircraft uses a **symmetric** section, and "the upward trim of elevons makes the symmetric airfoil to have reflexed camber line. It makes the positive pitching moment at the aerodynamic center" [51]. The reflex that trims that aircraft is not a property of its section but a deflected control surface held permanently out of line — and its authors note that the elevons are sized larger than a conventional flying wing's in order to produce enough moment for the transition. That is the same trade this configuration faces, resolved with a control surface where this one resolves it with twist, and it carries the same kind of continuous cost.

The position this paper takes is therefore the conservative one, and it is a statement about evidence rather than about physics. The trim moment this configuration needs is $C_m = +0.056$. No measured section approaches it. Whether a physically realisable reflexed section or planform could supply it is unresolved, and this paper does not assume either way; it closes the chain with twist, which is the route for which a computed requirement and a costed penalty both exist.

Two costs of the reflex route are measured, and both bear on a tail-sitter specifically. Reflex reduced maximum lift by about twelve percent in the first family [45] and by about ten percent in the second [46] — and maximum lift is what a tail-sitter needs at the high-incidence end of its transition. The same measurements also found that although minimum profile drag fell slightly, "if the profile drag coefficients are compared at equal values of the lift coefficient, the normal airfoil will be seen to have the lower profile drag except at small values of the lift coefficient" [45].

The reflex route is not free either, and the same literature prices it. A separate blended-wing UAV that trims by reflex rather than by twist records that carrying reflex over a wide span "is not conducive to the improvement of overall lift-to-drag performance" [44]. Both roads to trim on a tailless configuration therefore cost cruise efficiency, which is the reading Section 3.5 places on the 4.3 percent charged there: it is the price of having no tail, not the price of having chosen the wrong way to do without one.

The requirement 2R212 was built to meet is the one this configuration has. A survey of sections assembled for flying-wing use states it in the same terms — "for trim flight, pitching moment coefficient at zero angle of attack, C_{m0} , must be positive" — and identifies the practical family as the Eppler 184, 186 and 387, FX 69-H-083, NACA M5 and M6, Selig 5010 and 5020, MH 60 and HS-522 [27]. **No numerical value is taken from that survey here.** Its tabulated coefficients are negative for all ten sections, which contradicts the positive- C_{m0} requirement the same paper states two pages earlier, and a quantity this study cannot reconcile is not a quantity it will cite. The only measured number in hand remains the +0.004 of 2R212.

The most heavily cited experimental compilation of low-Reynolds section data cannot settle the question either, and says so itself: the UIUC low-speed tunnel "does not provide pitching moment data," so every moment coefficient in that compilation was "determined computationally using either the Eppler, ISES or XFOIL code" [36]. What those computed values show, however, points the same way as the measurement. The only flying-wing section in the compilation — the slightly reflexed MH45, designed for that application — carries $C_{m,c/4} = -0.006$, and the single positive entry among forty-one sections is not a reflexed section at all but a five-percent-camber high-lift section designed under an explicit low-pitching-moment constraint, at $+0.004$. That computed result points the same way as the measurements tabulated above: across four independent sources — one computational compilation and three tunnel campaigns — the reflexed and flying-wing sections return quarter-chord moments at or below zero, and the single positive value this configuration could lean on rests on one measurement from 1933. **The conclusion above is strengthened rather than weakened: the reflex route is closed, and the twist is what trims this aircraft.**

How much a section could contribute, if one existed that contributed more, is worth stating as a sensitivity — provided it is read as what it is. The rows below are not candidate sections. They are the answer to "how much would have to come from somewhere else", and the measured evidence sits in the second row, with the whole measured record of the preceding table at or below it:

If a section supplied $C_{m0} =$	Washout still needed	Inviscid e	Cruise L/D	Measured precedent
0	9.2°	0.865	11.68	four sections, at -0.001 to -0.007
0.004	8.6°	0.875	11.73	one section, NACA 2R212
0.020	6.0°	0.937	12.01	none
0.050	1.3°	0.986	12.21	none

At the one measured positive value, reflex buys half a degree of the nine and the trim problem stays a twist problem. The two lower rows of the table describe a section nobody has published a measurement of, and the four sections of the 4400R family fall *below* the top row rather than between the rows. **This paper therefore takes the branch the evidence supports** — nine degrees of washout and the penalty that goes with it — and does not offer the third row as an alternative design point. What would change this is a measurement, not an assumption: a reflexed section, tested in a tunnel that measures moment, returning a positive C_{m0} an order of magnitude above 0.004. Section 4 records that no such measurement was found.

What trims it is washout, which is how tailless aircraft have always been trimmed: on a swept wing the tips lie well aft, so negative tip incidence produces a nose-up moment about the centre of gravity. Because this mechanism is geometric rather than sectional, the vortex-lattice model of Section 3.10 — whose sections are symmetric — can compute it directly. Applying a linear twist from zero at the root to θ_{tip} at the tip and re-trimming to $C_L = 0.45$ at each value:

Tip washout	Trim α	C _m about CG	Induced C _D	Span efficiency e	Cruise L/D
0°	6.68°	−0.058	0.01077	0.993	12.65
−4°	8.20°	−0.034	0.01102	0.971	12.56
−6°	8.98°	−0.021	0.01141	0.938	12.43
−9°	10.16°	−0.001	0.01237	0.865	12.11

Nine degrees of tip washout trims the aircraft at cruise with no camber at all. That is a large twist by transport-aircraft standards and an ordinary one for a swept tailless design, and it is not free: the span efficiency falls from 0.993 to 0.865 and the cruise lift-to-drag ratio from 12.65 to 12.11, a **4.3 percent penalty paid to be tailless**. It is the same kind of payment the rest of this paper is about — a capability bought in one currency and charged in another — and it had not previously been counted.

One assumption is put at risk by this, and the risk runs the wrong way. Section 3.7 assumed an Oswald span efficiency of 0.85, and the range figures of Section 3.8 rest on it. Section 3.10 supported that assumption by computing an inviscid span efficiency of 0.99 for the planform and noting that an Oswald-type value runs at roughly 85 to 90 percent of the inviscid one, which places 0.85 at the bottom of the implied band of 0.84 to 0.89 — conservative, if only just.

That argument was made for a wing without twist. The wing without twist cannot be trimmed. For the trimmed wing the inviscid figure is not 0.99 but **0.865**, and the same reasoning then implies an Oswald value between **0.735 and 0.78** — a band the assumed 0.85 sits *above* rather than inside. If that reasoning holds, the cruise lift-to-drag ratio is 11.4 to 11.7 rather than 12.0, and the range figures of Section 3.8 fall by three to five percent.

That ratio has since been computed rather than assumed, and it is worth setting out what it replaced. An earlier version of this section could only bound the viscous-to-inviscid ratio from the literature. Traub reports both an inviscid Oswald factor from a vortex-lattice solution and a measured one from the tunnel for three wings [18]; the planar case gives a measured-to-inviscid ratio of 0.95, but his two non-planar cases give 1.01 and 1.07, which cannot be right for a real wing and are presumably an artefact of comparing an uncorrected tunnel measurement with an inviscid calculation. That data bounds the ratio loosely rather than fixing it, and the section carried a band of 0.85 to 0.95 with an implied penalty of one to five percent.

Section 3.10 now computes it on this planform, by calling the section solver at each strip's own local lift coefficient and integrating the profile drag across the span. The result:

	Inviscid e	Oswald e	Ratio	Cruise L/D
Untwisted	0.990	0.931	0.940	12.40
Trimmed, −9°	0.859	0.817	0.951	11.87

The direction the section predicted is confirmed and the magnitude is at the mild end of the band it allowed. The ratio is 0.951 rather than the 0.85 the borrowed rule

would have given, so the viscous penalty is smaller than feared; but the trimmed wing starts at 0.859, and the product lands at 0.817 — **below the assumed 0.85 by 3.9 percent**, costing 1.4 percent of cruise lift-to-drag ratio rather than the three to five percent this section had allowed. The range figures of Section 3.8 fall by the same 1.4 percent, and are not restated at the lower value because they are computed on the stated assumption; the correction is reported here instead.

Two things the computed number does not settle. Its sections are symmetric, so it is a lower bound: a cambered section reaching the same local lift coefficient at lower incidence would carry less profile drag. And the strip integration ignores sweep, which at 45° at the root is not a small omission — though the alternative, simple-sweep theory, halves the profile drag, which is a sign that the transformation does not apply to skin friction rather than a measure of the uncertainty. It is worth adding that 0.85 at this aspect ratio was a conventional choice rather than a careless one — an independent eVTOL sizing study adopts the same value at aspect ratio 7.0 [26] — which is precisely why it survived so long without being examined.

What is still not established. The twist here is linear, chosen for simplicity rather than optimised; a distribution shaped for span loading would trim at the same moment for a smaller efficiency penalty, so 4.3 percent should be read as an upper bound on the cost rather than as the cost. The vortex-lattice solution is inviscid, so it says nothing about how washout of this magnitude changes the stall behaviour of the outboard sections — washout normally improves it, which is a reason to expect no unpleasant surprise, not a demonstration that there is none. Two places where this result could have propagated were checked and do not. The root bending moment of Section 3.11, 934 N·m, was computed on an untwisted loading; washout moves lift inboard and can therefore only reduce it, so the spar sizing and the mass budget that follows are conservative rather than threatened — the recomputation has not been done because its direction is not in doubt. And washout makes the root stall before the tip on a swept wing, which is the favourable direction for the transition of Section 3.15 rather than the unfavourable one — a point that matters more than it first appears, because the tailless literature reports the unfavourable direction as a real hazard. The NASA series of 60° -swept flying wings found a pitch-up at high incidence that "became more severe as aspect ratio was increased", severe enough on some configurations to produce a *hung stall* — a trim condition with insufficient nose-down control to recover [20]. Those wings are of aspect ratio 1.15 to 2.15 and their pitch-up is driven by leading-edge vortex breakdown, which a wing of aspect ratio 6.03 swept at 45° at the root and 35° at the tip is unlikely to reproduce in the same form. But the trend they report runs the wrong way for a higher-aspect-ratio wing, and the incidences at which it appears overlap the 17° to 22° this aircraft passes through in transition. **This is a reason to keep transition controllability open rather than a new closure**, and it is the second independent reason to want the washout that Section 3.17 already requires for trim.

A second pitch-up mechanism was found afterwards, and it does not care about aspect ratio at all. In a blended-wing-body UAV analysed by both a low-fidelity method and RANS, the moment prediction departs from the computation above eight degrees of incidence because "the main wing stalls before the main body, causing the BWB to pitch-up" [39].

That is a statement about which part of the planform separates first, not about leading-edge vortex breakdown, so the aspect-ratio argument that limits the previous hazard offers no protection against this one. **Washout does.** On a swept wing washout makes the root stall before the tip, and on a blended wing the root is the body — so the twist Section 3.17 requires for trim also reverses the stall order that produces this pitch-up. The twist therefore earns its cost three times: it closes the trim chain, it delays tip stall on a swept planform, and it inverts the stall sequence behind the blended-wing pitch-break. **None of this makes the 4.3 percent penalty of Section 3.5 disappear. It means the penalty is not being paid for one thing.** And the aft limit of the centre-of-gravity window, roughly 85 percent of root chord, is still set by static margin alone and is still firm. What has changed is that the forward limit is no longer set by a section property nobody had measured; it is set by how much twist the design is willing to pay for.

The requirement is therefore smaller and more recognisable than first stated. An earlier version of this section asked for pitching-moment coefficients through ninety degrees of incidence, which would have needed a wind tunnel or an unsteady computational campaign. What is actually required is (a) the pitching moment up to roughly twenty-two degrees at low dynamic pressure, a mildly post-stall regime much closer to available data, and (b) trim at cruise incidence, which the configuration needs in any case. Neither is supplied here, and transition controllability remains the study's principal open item — but it is now an open item of ordinary size, and the reduced computational campaigns rejected earlier were rejected for substituting a less-validated model for a dominant term, not because the term is beyond reach.

One mechanism was omitted, and including it makes the case better rather than worse. The incidences above are geometric: the angle between the body axis and the velocity vector. But the nose propeller's slipstream passes over the inboard part of the wing — half its area on the light design and 63 percent on the heavy one — and inside that slipstream the air is not moving at freestream speed or in the freestream direction. Because the wake is aligned with the body axis, the resultant flow over the washed portion is pulled towards the body axis, and the local incidence there is smaller than the geometric one. Folk [23] writes this as

$$|V_a| = \sqrt{ |V_w|^2 + |V_i|^2 + 2 |V_i| |V_w| \cos \alpha }, \quad \alpha_e = \arcsin(|V_i| \sin \alpha / |V_a|)$$

with V_w the wake speed and V_i the freestream. Taking V_w from momentum theory at the transition thrust, and reporting both the fully developed wake ($V_w = 2v$) and the not-yet-developed case ($V_w = v$) because this study does not determine which applies at the wing station:

	Geometric α	$\alpha_e, V_w = v$	$\alpha_e, V_w = 2v$
Light, from a 5 m s^{-1} climb	17.5°	6.8°	4.2°
Light, from rest	21.6°	3.7°	2.0°
Heavy, from rest	20.5°	7.6°	4.6°
Heavy, from a climb	5.4°	4.7°	4.2°

Half the wing is therefore not post-stall at the moment of peak incidence, on either bracket. It is at four to eight degrees, which is ordinary attached flow. The pitching-moment budgets of this section were computed as though the whole wing saw the geometric incidence, so they are conservative — and the outstanding measurement, which Section 4 keeps open, is smaller than it appeared: it concerns the *outboard* half of the wing, the part that is not washed. Two cautions belong with this. Folk records that the reduced-order wake model overpredicts wake velocity above roughly 8 m s^{-1} , which places three of the four cases inside its stated range and the fourth — the heavy design entering from a climb — outside it, where the geometric incidence is only 5.4° in any case. And the verification of that model is attributed there to work the present authors have not read.

What the flight-test literature does about the same term, and what it found. The aerodynamic pitching moment through transition is not an item this study is alone in leaving open. A transition-optimisation study with outdoor flight trials builds its aerodynamic model by fitting published two-dimensional NACA 0012 lift and drag data through the incidence range, notes that its own aerofoil "is not exactly the same" and judges the error acceptable for a model-based controller, and carries **no pitching-moment term at all** in its equations of motion [28]. A second study does carry one, but as a linear form $C_m = C_{m0} + C_{m\alpha} \alpha + C_{m\delta} \delta$ used throughout the manoeuvre [29]. The practice of the field, in other words, is a borrowed section polar and a closed loop — which is what Section 3.15 does for lift and drag, and rather less than this section does for the moment. **The open item here is the field's open item.**

A third study carries the term in its fullest published form, and the form is instructive. Its equations of motion carry a nonlinear $C_m(\alpha)$ rather than a linear one, take the curve from a separate published study of the same vehicle rather than generating it, and evaluate the control-surface moment at "the sum of airspeed and slipstream speed near the quarter chord", closing the remaining discrepancy with an adaptive neural term rather than with better aerodynamics [42]. So the nonlinear moment does appear in this literature — borrowed, not derived, and backed by an adaptive law. Independently, a guidance architecture for the same manoeuvre computes an effective angle of attack from rotor wake velocity using hovering momentum theory [43], which is the same construction Section 3.15 applies above and a more recent statement of it than the source used there.

The pattern across all four is the same and it is worth naming. Every one of them obtains the transition aerodynamics by borrowing, fitting or adapting, and none by measuring the vehicle it flies. The measurement this paper leaves open is not one the field has made and this study skipped.

There is a reason nobody computes it, and it has been measured. A small blended-wing-body UAV was analysed by Reynolds-averaged computation and then tested in a wind tunnel at a Reynolds number of 2.0×10^6 . From -6° to 10° of incidence "both the aerodynamic force values and the variation trends are in quite good agreement"; from 10° to 26° they "show remarkable differences between the numerical and experimental results", which the authors attribute to the demands separated flow places on the computation [44]. The same boundary appears in the two other sources used here: a vortex-lattice solution of this class of configuration departs from computation above eight degrees [39], and the low-aspect-ratio flying-wing series records that its pitch-up is a separated-flow phenomenon [20]. **Three methods of three different fidelities fail at the same place, and the highest of them fails against measurement.** The incidences this aircraft passes through in transition — 17.5° to 21.6° geometric — lie inside that band.

This changes what the open item is. It is not that this study declined to compute the transition pitching moment; it is that the computation is not currently reliable for anyone at those incidences, on this class of configuration, and the literature says so with a wind tunnel behind it. **The item belongs to measurement, and Section 4 asks for it as measurement.** One encouraging observation travels with it: the same tests found the blended-wing configuration to have "soft-stall performance", which is the benign end of the range of behaviours the pitch-up literature describes.

That second study is worth more attention, because it flight-tested the question this configuration asks. Its authors compared their tail-sitter with and without elevons. Without them — that is, with attitude control by propeller differential thrust alone, as here — they report "a well-controlled attitude response during hovering and transition", but find that manoeuvres in level flight "cause an oscillatory attitude response" with "motor saturations observed", and conclude that the cause is "the increased aerodynamic moment but decreased motor thrust at high-speed level flight" [29].

That is the same structural tension this section derived from the moment budget, observed in flight. The tight case is not the high-incidence middle of the rotation but its end and beyond, where speed is high: the aerodynamic moment grows as V^2 while propeller thrust falls. The correspondence should not be pushed too far — their pitch propellers are the vehicle's lift rotors, mounted on the wing with a short arm, where these are dedicated control propellers on 0.71 m frames, and their vehicle is smaller — but the mechanism is identical and it saturated a real aircraft. **And it is not an isolated observation.** A survey of tail-sitter development records the same outcome for a separate vehicle, the CRC-20 developed at the University of Maryland with wind-tunnel-derived aerodynamics: its builders "were able to achieve transition to forward flight, but they had poor control over the vehicle once in forward flight" [30]. Two independent programmes, different vehicles, the same division — transition passed, forward flight difficult.

That is the strongest external reason to treat the end-of-rotation budgets of 0.050 and 0.010 as the binding numbers in this section, and the strongest external argument that a propeller-only tail-sitter needs its pitch authority checked at speed rather than at incidence. It also says something about where a future experiment should be pointed: the

regime that has caused trouble in practice is not the one this paper spent the most effort on.

Two limits of this reading are worth stating. The incidence history comes from the point-mass trajectory of Section 3.15: it is the geometric angle between the body axis and the velocity vector, so it is only as good as that trajectory. And the rotation rate itself varies the local incidence along the body by $\omega \bar{c} / 2V$, which is $\pm 3.1^\circ$ for the light design entered in a climb and $\pm 8.6^\circ$ entered from rest — so at the peak-incidence instant of that second case parts of the airframe see close to thirty degrees. Nothing here should be read as a demonstration of transition authority.

Supplementary S5 — The limitations in full

Supplementary material to "The Architectural Cost of Hybrid VTOL: meryemAircraft, a Propeller-Driven Tail-Sitting Blended-Wing-Body Without a Dedicated Lift System".

This is the full version of Section 4 of the paper, reproduced here without abridgement. Section 4 states the limitations that bear on the conclusions; this material states all of them, item by item, with the break-even value of every assumption that has one and the reasoning behind each. It ends with the list of places where these results could most efficiently be attacked.

S5.1 No experimental validation

There is no wind-tunnel testing in this work and no flight testing. Nothing in Sections 2 to 3 has been measured against an experiment.

One coefficient has been computed rather than assumed or taken from the literature: the zero-lift drag of the wing and centre body, by a three-dimensional Reynolds-averaged solution reported in Section 3.10. That is a calculation, not a validation — it is subject to the turbulence-model uncertainty quantified there, which is its largest term, and it has not been checked against measurement. Every other aerodynamic coefficient is either taken from the literature or assumed.

S5.2 The mass budget is bounded from below, and the bound is thinner than it looks

The reference designs are sized from an assumed mass breakdown — 30 % structure, 16 % propulsion chain, 4 % battery, 8 % avionics, 16 % fuel, 26 % payload — and that breakdown was, in the first version of this study, a target rather than a finding. A component build-up has since been carried out and is reported in Section 3.11. It closes at 50 kg — the components sum to a payload residual of 30.4 percent against the 26 percent assumed, a margin of 2.2 kg — but it closes *conditionally*, and the conditions are worth stating as the result rather than as a footnote to it: **the light design closes if the**

average structural areal density stays at or below 1.78 kg m⁻², and if everything still outside the model together stays below 2.2 kg. That converts the assumption from an assertion into a bounded claim. It does not make the claim comfortable, for three reasons.

The margin lives in one number. Breaking each assumption in turn to find the value at which 13 kg of payload no longer closes gives margins of 38 to 197 percent on the propulsion and assembly terms, and 19 percent on the shell areal density. At 1.78 kg m⁻² of skin rather than the 1.5 assumed, the payload is gone.

The 1.5 is no longer without an anchor, though the anchor is not a direct one. A doctoral study of unmanned-aircraft sizing reports an areal density of **1.05 kg m⁻²** for the fuselage of a small UAV built as a monolithic composite shell supported by an internal structure [50]. That is below the figure assumed here, so the assumption is conservative against it — but the comparison is not like for like: that shell is carried by an internal structure, while the shell here is the wing and carries flight loads directly. What the figure establishes is that the assumed value is not outside the range small composite airframes are built to. It does not establish that this airframe can be built to it. Every other line could be substantially worse than assumed and the design would still close; that one line could not.

The build-up came in lighter than the target, which is a warning and not a reassurance. Its structure is 23.8 percent against the 30 assumed. The first pass of the same build-up returned a payload fraction of 42.8 percent, and the difference between that and the 30.4 reported is seven categories of hardware that the first pass had simply omitted. A build-up that has already been found to be missing 3.4 kg of items may still be missing more. What that risk costs is quantifiable, and it is exactly the margin already stated and not an additional allowance: the contingency line can rise from the 12 percent of dry mass used to 22 percent before the payload claim fails, which is 2.68 kg growing to 4.89 kg — a further 2.2 kg of unaccounted mass, and no more. An earlier version of this section quoted 4.5 kg by reading the break-even contingency as if it were additional to the budget rather than inclusive of what is already in it. The two figures were never independent: the survivable unaccounted mass and the payload margin are the same 2.2 kg, counted once.

The four categories most often missing from a build-up of this kind were checked explicitly and are present: propeller blades and coaxial hubs, power electronics sized on hover peak rather than on engine rating, fuel containment as distinct from fuel, and control actuation — the last being small here only because the configuration has no control surfaces to actuate. What is *not* separately modelled is local load introduction: the tip-frame roots, the nose motor mount, and the payload, fuel and battery supports are carried inside the shell and internal-structure allowances rather than sized. If those load paths cost more than the allowances contain, they come out of the same 2.2 kg.

Paper aircraft are habitually lighter than the aircraft that eventually get built, and the payload fraction remains the number most exposed to that, because payload is the residual and absorbs the entire error of every other line. This is still the single most likely place for the results of this paper to be wrong, and it remains the reason Section 4.1

declines to compare the calculated payload fractions against the published figures of aircraft that exist.

One structural question is narrowed by the build-up rather than settled. At 50 kg the root bending moment is 934 N·m, which a carbon spar cap of 10.7 mm² carries at the design allowable; the caps weigh 41 grams, eight parts in ten thousand of take-off mass, and under one percent even at 1000 kg. **Global span bending is therefore not the sizing driver in this model** — which is why the twenty-five percent thick centre body costs nothing in bending. That is the whole of the claim. Buckling, core shear, torsion, local load introduction at the tip-frame roots and the nose mount, minimum manufacturing gauge, damage tolerance and aeroelastic margin are outside the model, and an earlier version of this section over-read the 41 grams as showing that strength in general is not the driver. It does not. Those modes are carried, if at all, inside the shell areal density and the internal-structure allowance — which is a further reason the shell figure is the number this budget stands or falls on.

S5.3 Geometry chosen rather than derived

The sweep distribution — 45° at the root falling to 38.3° at the tip on the leading edge, with the trailing edge constant at 25° — was selected from a 20°–40° band reported favourable in the transonic-transport literature. The present aircraft is subsonic. The band has not been re-derived for this flight regime, and the values are therefore design choices supported by precedent rather than results. The realised sweep variation is under seven degrees, which is smaller than the crescent-wing precedent of Section 1.3 would suggest; the planform inherits the *principle* of a coupled sweep-chord-thickness distribution, not the magnitude of the original.

The taper distribution has likewise not been optimised, and the aerofoil sections are described by thickness, camber and reflex distributions rather than by specific sections.

S5.4 Coefficients taken from the literature

Several results depend on coefficients that were not computed for this geometry:

- The **roll authority** of the lower-surface strip. Section 2.10 now computes the two halves of this that can be computed for this geometry — the roll inertia, 25.0 kg·m², and the roll damping, $|C_{l_p}| = 0.358$ from a vortex-lattice solution rather than from the literature — and inverts the question: twenty degrees per second at cruise requires 27.1 N·m. What remains from the literature is the strip's own effectiveness, and it is the load-bearing part. The strip's own force as a swept fence supplies about a third of the required moment; the moment must therefore come from the change in the half-wing's circulation, which asks for $\Delta C_L \approx 0.12$ over the strip's span. Published wind-tunnel data for a two-percent-chord device over the same inboard two-thirds of span report increments of 0.14 to 0.17 [18], so the requirement is below what a comparable device delivers — but that device is a trailing-edge flap on an unswept wing, and **roll authority remains sized and not closed**. The same comparison shows the strip's height law to be wrong outboard, reaching 13.7

percent of local chord where the mechanism saturates near two; Section 2.10 records this as a change the configuration needs. An earlier version of this paper quoted 46 N·m without stating the mechanism it came from; that number implies $\Delta C_L \approx 0.20$ and is not reproduced here as an authority.

- The **directional stability** of the configuration. Section 2.10 computes $C_{n_\beta} = 0$ for the planform — the wing supplies none, which measured data on tailless aircraft confirm is normal rather than unusual [19,20] — and shows that the tip-frame fairing must supply it, at a chord well inside what the fairing needs for drag reasons. Two items are added by that literature and closed by neither it nor this paper: the fairing needs a **toe angle**, whose sign follows from its aspect ratio and which brings a stall-related failure mode with it; and the fin's **drag** contributes to directional stability at an arm of half the span, a mechanism the estimate here omits and which makes it conservative. The fin contribution itself is **not computed**, and neither is the yaw damping it would bring; the profile-drag damping of the bare planform, $C_{n_r} = -0.0023$, is negligible. Yaw authority is not in question — the yaw arm is the semi-span, so the available moment is 2.4 times the pitch moment — but directional stability and yaw damping are a single open item resting on a component whose section has not been selected.
- The **battery buffer's specific power**, which Section 3.11 now identifies as the most exposed number in the mass budget. Taken at the electrical bus, where the buffer is, it asks for **5.63 kW kg⁻¹** to hover and 6.48 to leave the ground, where the only figures this study has read give 0.7 to 1.3 kW kg⁻¹ at pack level for power-application Li-ion [21]. At the top of that range the buffer would mass 7.8 kg instead of 1.8 and **the light design's budget would not close**. The assumption is that a purpose-built short-duration buffer beats an automotive traction pack by four to nine times; that is plausible in kind and unverified in magnitude, and Bill 3 rests on it. The energy side does *not* compound it: the buffer discharges through a take-off of ten to twenty seconds rather than continuously, and even a low-energy high-power cell would use only a third of its capacity doing so. Power is the binding currency, and a shortfall in it is a floor rather than a transient — thrust goes as power to the two-thirds, so a buffer at 1.3 kW kg⁻¹ gives a thrust-to-weight ratio of 0.71 and the aircraft does not leave the ground.
- The **cruise state of the tip rotors**. Section 3.3 shows that they must turn freely at zero shaft load rather than stop, on pain of adding 61 to 74 percent to the zero-lift drag, and that fixed pitch removes feathering as a third option. The estimate behind that is an area-and-coefficient calculation with assumed solidity and section drag coefficients, not a propeller calculation; what is robust is the ratio between the states, not the values. It also converts a seized tip-rotor bearing from a control failure into a drag failure, and no reliability analysis of that is offered here.
- The **three-axis coupling of the roll strip**, none of which was previously stated. Deploying it on one side produces adverse yaw of 2.9 to 11.3 N·m depending on the height law, against 42.8 to 55.9 N·m of yaw authority — covered. It also produces a nose-down pitching moment, measured for this class of device [33] and bracketed here at ΔC_m of 0.005 to 0.032 against a tightest budget of 0.050 — **not comfortably covered at the upper end**, which yields a scheduling restriction

rather than a redesign. Both estimates carry the same unselected-cross-section caveat as the frames, and the chordwise position of the lift increment, which sets the pitch coupling, is not computed anywhere in this paper.

- **Turbulence and post-stall degradation of the strip.** The mechanism the roll requirement depends on is reported to weaken past the stall angle and to become negligible at 19 percent inflow turbulence [35]. The strip's inboard portion sits in a propeller slipstream and the transition passes through post-stall incidence, so both conditions are met somewhere in the flight envelope. The cited measurements are on a different aerofoil in grid turbulence and transfer only in direction; no measurement of this strip in a slipstream exists. **This is the principal risk to hover roll authority**, and it sits on the same claim — slipstream placement — that Section 2.10 offers as an advantage. The post-stall half of this item is contested rather than established: a simulation at twenty degrees of incidence returns a ninety-four percent lift increase from a flap of the same height fraction at essentially unchanged drag [37]. The turbulence half is unopposed. Against both, a water-tunnel study at a Reynolds number of 8 588 found the mechanism qualitatively unchanged four orders of magnitude below the usual test range [38], which is evidence that low speed as such is not the threat — turbulence is.
- The **roll actuator**. The strip extends by a commanded amount, so the limit cycle computed in Section 2.10 — $\pm 0.2^\circ$ in bank at a fifty-millisecond deployment, $\pm 9.4^\circ$ at a hundred and fifty — bounds the two-position worst case rather than describing normal operation. What is not stated anywhere is the actuator's mass, its power draw or its bandwidth, and no closed-loop stability analysis has been carried out.
- The **frame drag** of Section 3.3 uses C_D values representative of circular and faired sections at the relevant Reynolds number. **The frame cross-section has not been selected.** The conclusion that the frames must be faired is robust — the difference between $C_D = 1.15$ and $C_D = 0.15$ is not a matter of coefficient precision — but the twelve-percent figure is an estimate.
- The **zero-lift drag coefficient** of 0.0248 is assumed rather than adopted from a build-up. Section 3.10 reports a component build-up that brackets it at 0.0131 to 0.0210, so the assumed value is conservative. That build-up had a weak link in its largest term — its strip method treated the root section as a two-dimensional aerofoil of twenty-five percent thickness, and the flow over the centre body of a blended-wing body is not two-dimensional. **That link has since been replaced** by a three-dimensional solution, also reported in Section 3.10, which brackets the wing-and-body term between 0.0120 and 0.0148 depending on the turbulence closure and on the starting field, and the total between 0.0201 and 0.0231 — still below the assumed value in every case. What remains uncertain is no longer the dimensionality but, first, the transition state — the solution is fully turbulent, and the clean-surface case is still the strip estimate — and, second, the uniqueness of the solution itself, since the wall-resolved SST case settles four percent apart from two different starting fields. The build-up is reported as a bound on the assumption rather than as a replacement for it.
- **Span efficiency** is assumed at 0.85, and the assumption is now bounded by a calculation rather than by a borrowed rule. A vortex-lattice solution gives an inviscid

span efficiency of 0.99 for the untwisted planform and 0.859 for the wing twisted to trim. Section 3.10 computes the corresponding Oswald-type values by calling a section solver at each spanwise station's own local lift coefficient: **0.931 untwisted and 0.817 trimmed**, a viscous ratio of 0.94 to 0.95. The assumed 0.85 is therefore **optimistic by 3.9 percent**, worth 1.4 percent of cruise lift-to-drag ratio and of the range figures of Section 3.8. What the calculation does not settle: its sections are symmetric, so it is a lower bound rather than an estimate; it ignores sweep, and the root sweep is 45°; and it inherits whatever magnitude error the vortex-lattice solution carries on a configuration of this class, which Section 3.10 reports as untested and bounded above by a published comparison.

S5.5 Torque balance holds at one point only

The propeller pairs are of fixed geometry, so exact torque cancellation occurs at one operating condition. That condition was chosen to be cruise, on the grounds that cruise is long and strategic while hover is short and tactical. A small residual torque therefore remains in hover. It is trimmed by the same lower-surface strip that provides roll in cruise, but the trim authority required has not been computed.

S5.6 The transition simulation is a point mass

The results of Section 3.15 come from a two-degree-of-freedom point-mass simulation in which the body angle is driven kinematically. It therefore does **not** model rotational dynamics, and the tip-propeller thrust required to produce the rotation does not follow from it. Section 3.17 supplies part of what is missing — the inertia about the rotation axis, the peak angular acceleration a finite moment can actually produce, and the resulting margin — but only part: the aerodynamic pitching moment through ninety degrees of incidence is still absent, so what that section establishes is a necessary condition and not a sufficient one. It also records that the linear angle ramp used in Section 3.15 cannot be produced by any finite moment, and reports the measured sensitivity of the altitude-loss tables to that choice.

One published number changed as a result. The heavy reference design's rotation time was four seconds, at which — against the corrected inertia of Section 3.17 — the margins are 0.97 on the cheapest profile and 0.65 on a smooth one, that is, infeasible on both. It is now 5.1 s, the time at which the heavy design holds the light design's margins, and the change costs nothing: the tip-propeller power falls from thirteen percent of hover power to six, and the altitude-loss result is unchanged. The light design is on the same boundary rather than clear of it: two seconds against a smooth minimum of 2.01 s. Both reference rotation times should be read as actuator-limited lower bounds. The aerodynamic model is a linear lift curve to stall with a flat-plate relation beyond it; dynamic stall, separation hysteresis and propeller-wake effects on the wing are absent.

The qualitative conclusion — that a slower rotation loses less altitude, and that entering the rotation while climbing removes the loss entirely — depends on the sign of the vertical support term rather than on the details of the aerodynamic model, and is robust. The specific altitude figures are not.

That robustness has since been tested rather than asserted. The simulation takes its lift-curve slope from the thin-aerofoil expression, 4.72 rad^{-1} ; the vortex-lattice solution of Section 3.10 gives 3.87 rad^{-1} for this planform — eighteen percent lower, and in the direction that would make the aircraft fall further. Repeating both tables with the lower value moves no published entry by more than 1.2 m, and the two reference profiles — two seconds for the light design, 5.1 for the heavy, entered at 5 m s^{-1} of climb — still lose no altitude at either slope. The conclusion of Section 3.15 survives an eighteen-percent error in the coefficient it rests on.

S5.7 The tip-surface benefit is not quantified

The tip frames extend perpendicular to the planform by 0.41 of the semi-span, which is four to eight times the relative height of a conventional winglet. Giving their fairings a lifting section rather than a symmetric one costs no additional part, mass or mechanism, and induced drag is 33.7 % of cruise drag at the reference condition, so the mechanism has something to act on.

No number is claimed for it. The surface is far outside the geometric range for which textbook winglet relations were established, and estimating the benefit properly requires a panel method or CFD. The associated costs — increased root bending moment, and increased directional stability that the tip propellers must overcome to command yaw — have likewise not been quantified.

S5.8 Disturbance rejection in hover

With the transition no longer sizing the tip propellers, hover disturbance rejection becomes the sizing case. Only an order-of-magnitude check has been made against a single gust condition. The disturbance spectrum, the closed-loop bandwidth required, and the actuator response needed to achieve it have not been analysed.

S5.9 Ground handling and crosswind

No claim is made that the aircraft resists tipping in arbitrary ground wind, and the historical record gives a specific reason not to make one: the contemporary assessment of the XFY-1 quoted in Section 1.2 records "tip-over tendencies noted when on ground in gusty air" [1]. This is a property of standing an aircraft on its tail and it is inherited here. The stance base is a design parameter that can be widened without altering the configuration, and the reference geometry represents one point on that trade rather than a limit. Operating limits in ground wind are an operational matter, and published data for this class shows that such limits are ordinary rather than exceptional: a fielded fixed-wing VTOL uncrewed aircraft in the same mass range quotes a wind limit of 15 knots for take-off and landing against 25 knots in cruise [8]. A lower ground-wind limit than cruise limit is the normal condition for VTOL aircraft, not a defect peculiar to tail-sitters. The specific limits for this configuration have not been computed here.

S5.10 Vertical descent

The vertical descent has not been analysed. A rotor descending into its own wake can enter the vortex ring state, in which thrust becomes erratic and additional power is counterproductive. Whether and at what descent rate this configuration encounters that region is an open question and one of the more important items of future work, because it bears directly on the landing phase.

S5.11 Prior art

The novelty claimed in this paper is a combination, and the elements of that combination individually have antecedents. Blended-wing bodies, tail-sitters, coaxial counter-rotating pairs, series-hybrid powertrains and attitude control by differential thrust have all appeared before, separately and in various partial groupings. In particular, full attitude control of a tail-sitter with no control surfaces at all has been demonstrated and published; that aircraft differs from the present configuration in its planform, in distributing thrust across the span, and — decisively — in taking its rolling moment from the differential reaction torque that the coaxial arrangement here removes by design.

The literature underpinning Sections 1 and 2 was read at first hand where the sources could be obtained. Seven were: the two NASA reviews of United States V/STOL development on which Section 1.2 rests [1,2], the doctoral study from which the drag measurements of Section 2.3 are taken [3], the QuadPlane wind-tunnel characterisation [4], the stationary-lift-propeller drag study [5], the concept-vehicle sizing study quoted in Section 2.2 [6], and the tail-sitter flight-test paper cited in Sections 2.4 and 2.10 [7]. Three further sources were sought and not obtained — the journal version of [3], an earlier conference paper by the authors of [7], and a 2025 forum paper on stopped-rotor drag — and none of them is relied upon for any claim here; where a claim had rested on the last of these, it was removed rather than retained on a summary. No patent claim text was read in the original; the prior-art position stated here is that of an author survey, not of a professional search.

S5.12 The architecture comparison is conditional, and on two things rather than one

The comparative sizing of Section 3.6 settles the case against a separate lift system using measurements, and does not settle the case against a tilting mechanism at all. Two separate conditions carry that second result and both should be read as limitations.

The first is an unmeasured number. The tilting layout is credited with paying no cruise drag for its nacelles, pivots and hover-pitched blades, because no measurement of that penalty was available to charge it with; on that credit it cruises twelve percent further than the configuration proposed here under the fixed-fuel-fraction rule. The sign of that comparison reverses if the penalty exceeds roughly eleven percent of cruise drag, which is what the tip frames of this configuration cost it. The reader should treat the tilting column as an upper bound on that architecture rather than as an estimate of it.

The second is the sizing rule itself, and it applies symmetrically to every architecture including the one proposed here. Range as computed from a fixed fuel fraction does not depend on take-off mass, so an architecture that closes heavier is permitted to carry proportionally more fuel and its mass penalty never reaches the range column. This is a correct property of the Breguet form and not an error, but it is a poor contract for comparing architectures, and a comparison reported under it alone would be misleading. Section 3.6 therefore reports three contracts — fixed fuel fraction, fixed fuel mass, and fixed take-off mass with fixed payload — and the tilting layout's advantage survives only the first. Which of the three is the right question depends on what is being procured: a fixed mission, a fixed fuel load, or a fixed vehicle class. This paper does not choose among them, and no result here should be quoted without the contract it was computed under.

Three further caveats sit under the same model. Wing loading and disc loading are held common across the three architectures, which is a controlled comparison and not a statement that each is at its own optimum. The structural fraction is likewise common, which is generous to the distributed-lift layout, since carrying power to the extremities is generally held to carry a structural penalty of its own; charging it makes that layout heavier without changing its range, so the direction of the result is unaffected. And the second table in that section, which gives each architecture its own unbuffered power system, is a bounding case rather than a fair comparison, for the reason given there.

S5.13 The engine rating margin is not consistent between the two reference designs

The sizing model of Section 3.6, calibrated entirely on the light reference design, predicts the heavy one to within four percent in take-off mass and one tenth of a percent in range without a single coefficient being changed. One term does not carry across. The engine is rated at 2.6 kW against 1.7 kW of cruise electrical power in the light design, a margin of 1.53, and at 54.3 kW against 39.2 kW in the heavy one, a margin of 1.385 — ten percent apart, and nowhere justified in this paper. Carrying the light margin through to the heavy design overstates its engine by seventeen percent while leaving range and lift-to-drag ratio untouched. A larger generator and power electronics being relatively more efficient is a defensible reason for the difference, but it is a reason supplied after the fact; as the two designs stand, the margin is an undeclared choice rather than a scaling law, and the scale-invariance claimed in Section 3.9 should be read as holding for the mass and range fractions and not for this one.

S5.14 The strip's authority does not start at zero

Extension is the control variable, but a device of this kind produces no lift change until it projects about one percent of the local chord [19]. The strip is tapered, so the threshold is crossed progressively from the outboard end inward: it is wholly inert below seven percent of commanded travel, and above a quarter of travel the response is linear to within five percent. The intervening band is genuinely nonlinear and is not modelled anywhere in this paper. The consequence is confined to small corrections rather than to large ones, but a closed-loop design would have to carry it, and no closed-loop design is attempted here.

A second, larger omission sits beside it. The strip projects from one surface only, and the pitching moment that arrangement produces was described in the older literature as "prohibitive" for spoilers used as ailerons, with projection from both surfaces named as the remedy [19]. Section 2.10 resolves the coupling by restricting when the strip may be commanded. The alternative resolution — a strip that projects from both surfaces, so that the pitching contributions oppose and the rolling contributions add — has not been examined, costed or ruled out.

S5.15 The directional-stability estimate omits the body, and the sign of the omission is known

The vortex-lattice model used for C_{n_β} has no volume. The centre body of a blended wing develops side force in sideslip ahead of the centre of gravity and is therefore destabilising, and on tailless aircraft that contribution is reported to be "at least as great as the stabilising effects contributed by the wing alone" [19]. The planform contributes zero here, so that statement yields no magnitude — only a direction. The fairing sized in Section 2.10 is sized to bring a zero up to criterion; the real starting point is below zero, and the shortfall is unquantified.

The same section's fairing rests on an assumed lift-curve slope of 4 per radian at a chord Reynolds number near 80,000. Symmetric sections at low Reynolds number are measured to be nonlinear about zero incidence, in one case reversing the sign of the lift-curve slope over a three-degree band [36] — the same band a toe angle of one to two degrees occupies. Whether a fairing of that chord develops the side force credited to it is not established.

S5.16 The toe-out arrangement has a failure mode at large sideslip

The frames' aspect ratio requires toe-out (Section 2.10). Toe-out carries a known hazard: yawing far enough to stall the rear fin produces a large *destabilising* yawing moment, where the same stall on a toed-in fin produces a stabilising one [19]. The sideslip angle at which that occurs sets an upper bound on the usable sideslip envelope, and it has not been computed. It depends on the fairing's section and Reynolds number, neither of which is fixed here.

S5.17 What would change these conclusions

The results of this paper would be most efficiently attacked in six places, and they are listed so that they can be:

1. **~~A three-dimensional solution for the centre body.~~ Done.** This was the first place to attack, because the strip method of Section 3.10 could not model the flow over a twenty-five percent thick blended centre body. The solution has since been carried out and is reported in Section 3.10: it gives a wing-and-body C_{D0} of 0.01475 with the Spalart-Allmaras closure and 0.01201 – 0.01253 with $k-\omega$ SST — a spread of eighteen percent between the closures that nothing in the solutions resolves — and it leaves the assumed 0.0248 conservative in every case. The dominant term is the turbulence model, and it is larger than first reported because the two models were subsequently paired at the same wall resolution rather than at two

different ones. **What it does not settle** is, first, the transition state — the solution is fully turbulent, so the clean-surface figure of 0.0073 is untested and the gap between it and 0.0120–0.0148 is now the largest single uncertainty in the zero-lift drag, larger than the spread between the closures. A transition-sensitive model requires a wall-resolved grid and a two-equation formulation at once; that combination now converges, but the model could not be validated in the low-turbulence regime the cruise condition sits in. **Second, and newly opened, is the uniqueness of the wall-resolved solution itself.** That case cannot be started from a uniform field, and the two starts that do converge — one warmed from the Spalart-Allmaras solution, one mapped from the coarser SST solution — settle 4.3 percent apart, entirely in the pressure component. Steady RANS is admitting more than one stationary solution here. **This item is therefore narrowed rather than closed**, and what replaces it is stated above.

2. ~~**A structural mass estimate** for the airframe and the tip frames.~~ **Attempted; conditional at 50 kg, open at 1000 kg.** The build-up of Section 3.11 meets the 50 kg payload fraction with 2.2 kg in hand, on the condition that the shell areal density does not exceed 1.78 kg m^{-2} and that everything still outside the model together stays under that same 2.2 kg. **What it does not settle** is either of those conditions, or how the areal density scales: the 1000 kg design is at break-even when it grows as the 0.467 power of linear scale, and that exponent was not measured. The build-up also does not contain buckling, torsion, local load introduction or aeroelastic sizing, and its tip frames were sized for a vertical landing only. **This item is therefore narrowed rather than closed**, and what a fuller version of it would have to bound is now specific.
3. **A six-degree-of-freedom transition simulation** with rotational dynamics. **Partly done, and the remainder is blocked on data rather than on effort.** Section 3.17 derives the inertia from the component build-up and shows the tip propellers carry it with a margin of 1.49 at the light design point and 1.57 at the heavy one on the cheapest rotation profile, and of 0.99 and 1.05 on a smooth one — that is, at the actuator limit rather than clear of it — together with a measured account of how that margin narrows with size, and a threshold for the aerodynamic moment resolved along the trajectory. That resolution corrected the requirement rather than merely quantifying it: the aircraft does not reach ninety degrees of incidence, because the relative wind rotates with the body, and peak incidence is 17.5° to 21.6° . **Transition controllability remains the largest unresolved item in this study**, but it now asks for the pitching moment up to some twenty-two degrees at low dynamic pressure and for trim at cruise incidence, rather than for a moment sweep through ninety degrees — and Section 3.17 narrows it once more, since the inboard half of the wing lies in the nose propeller's slipstream and sees an effective incidence of four to eight degrees rather than the geometric seventeen to twenty-two [23]. The measurement that is still owed concerns the *outboard* half. Two further things are now known about this item. It is **the field's open item, not this study's alone**: published transition analyses with flight trials carry either no pitching-moment term or a linear one, and close the loop with a controller instead [28,29]. And a flight-tested tail-sitter controlled by differential propeller thrust alone, with no elevons, was well behaved through hover and transition but saturated its motors during manoeuvres in **level flight**, its authors attributing this to aerodynamic moment growing with speed while thrust falls [29]. **The binding condition for a propeller-only tail-sitter is therefore pitch authority at speed, not at incidence** — which is where the tightest budgets of Section 3.17, 0.050 and 0.010, already sit. What is still not done is answering it: a reduced computation would replace a dominant term with a less-validated model. What it cannot do is charge the aerodynamic pitching moment, which requires moment coefficients through ninety degrees of incidence; those are not available for this planform and cannot be produced without a wind tunnel or a dedicated computational campaign. **This item is therefore reduced to a specific missing measurement rather than a missing analysis.**

4. **A panel-method analysis of the tip surfaces**, which would either convert S5.7 into a quantified benefit or remove it. The vortex-lattice solution of Section 3.10 covers the planform but not the tip surfaces, which remain unquantified.
5. **~~A viscous solution of the twisted planform, section by station.~~ Done.** The Oswald-type efficiency the drag build-up needs was obtained from the inviscid figure by a borrowed ratio. It is now computed on this planform by the method the non-linear vortex-lattice literature uses — two-dimensional viscous analyses at each spanwise station coupled to the three-dimensional circulation [40] — and reported in Section 3.10. The borrowed ratio of 0.85 to 0.90 was too pessimistic; the computed ratio is 0.94 to 0.95. The conclusion nevertheless stands in the unfavourable direction, because the trimmed wing starts from a lower inviscid figure: the Oswald efficiency is **0.817 against the assumed 0.85**, worth 1.4 percent of cruise lift-to-drag ratio. **What it does not settle** is, first, that the sections used are symmetric, so the figure is a lower bound rather than an estimate; second, that strip integration ignores sweep, and the root sweep is 45°; and third, that it does nothing about the magnitude question of item 6 below, since it is built on the same vortex-lattice loading. **This item is therefore closed as to the ratio and open as to the absolute level.**
6. **A Reynolds-averaged or panel solution of this planform's loading**, to bound the magnitude error the vortex-lattice results carry. A published comparison on a blended-wing-body of this class found the vortex-lattice lift coefficient low by thirty to thirty-eight percent against RANS [39]. Section 3.10 argues that a near-constant multiplicative error of that kind cancels in the ratios this paper takes from the solution, but the check that would confirm it — whether the moment scales with the lift by the same factor — is withheld in that source. This is the one exposure in the aerodynamic chain with no bound at all.

Not all of these are within reach of a calculation, and an earlier version of this list said they were. The first has been carried out and its result is folded into Section 3.10, and so is the fifth. Of the remainder, the transition pitching moment is blocked on data rather than on effort, for the reason item 3 gives — three methods of three fidelities fail above roughly ten degrees of incidence — and the fin derivative of S5.15 is blocked the same way, because what the low-Reynolds-number measurements remove is the linearity of the curve a calculation would have to assume. Those two need a tunnel. The rest are within reach of a follow-on study, and the configuration is described in enough detail in Section 2 and Section 3 for another group to attempt any of them independently. The computational setup, the grid-convergence study and the record of what failed along the way are in the repository, so the first item can be re-run and checked rather than taken on trust.

Supplementary S6 — The three bills stated formally, and a comparative sizing

Supplementary material to "The Architectural Cost of Hybrid VTOL: meryemAircraft, a Propeller-Driven Tail-Sitting Blended-Wing-Body Without a Dedicated Lift System".

This material is Sections 3.1 and 3.6 of the paper, reproduced here in full. The first states the three bills as equations with the transfer table showing that they are one quantity in three currencies; the second sizes three architectures against the same mission under three different sizing contracts, all twelve cells, with the sensitivity sweeps.

S6.1 The three bills stated formally, and a test of the statement

The argument so far has been verbal. It is worth stating compactly, because the compact form makes clear what the framework claims and what it does not.

For an architecture a flying a given mission, write the three charges as fractions of the quantity each degrades:

$$f_1(a) = \frac{m_{\text{hover-only}}(a)}{\mathrm{MTOW}}, \quad f_2(a) = 1 - \frac{(L/D)_a}{(L/D)_{\text{clean}}}, \quad f_3(a) = \frac{P_{\text{cont}}(a) - P_{\text{cruise}}}{\sigma_P \mathrm{MTOW}}$$

where $m_{\text{hover-only}}$ is the mass that exists solely to hover, $(L/D)_{\text{clean}}$ is the lift-to-drag ratio the airframe would have with no hover hardware exposed, P_{cont} is the continuously installed power, and σ_P is the specific power of the power system. Each is dimensionless, each is zero for an aircraft that does not hover, and each is measurable for one that does.

The claim of Section 2.6 is that the same architectural choice need not minimise all three simultaneously. The architectural moves available typically move cost between them rather than removing it: retracting the lift rotors reduces f_2 and raises f_1 by the retraction mechanism; tilting the propulsors reduces f_1 and f_2 together and introduces a mechanism whose mass and failure modes are the price; buffering the hover peak reduces f_3 and raises f_1 by the buffer. Section 2.4 tabulates these transfers. The escape condition is the statement that all three vanish simultaneously only when the hover and cruise hardware are the same hardware, in the same orientation, doing the same job, with the peak supplied from a buffer.

A consequence that can be checked against published work. If the three are genuinely separate currencies rather than three names for one quantity, then an architecture may be *best* in one and *worst* in another — in particular, the architecture with the highest cruise lift-to-drag ratio need not be the lightest. A single-metric comparison would not anticipate that. The NASA sizing study quoted in Section 2.2 reports exactly this pattern: the lift-plus-cruise concepts are the heaviest of the four examined *while having the highest cruise efficiency of that group*, and the authors attribute the weight to hardware carried for hover rather than to cruise power. That is f_1 dominating while f_2 is favourable, which is the framework's prediction and not a restatement of it. The later and larger version of the same programme states the transfer in a single sentence: "the high cruise efficiency of the lift+cruise type reduces the battery weight compared to the quadrotor, but not enough to counter the increase in structure and propulsion weight, so the all-electric lift+cruise aircraft is the heaviest design" [22]. A gain in one currency, insufficient against a loss in another, named as such by authors with no framework to defend.

That later study also contains the case that would embarrass the framework if it behaved differently, and it does not. Sizing five architectures rather than four to the same mission adds a tiltwing, and the tiltwing has the highest cruise efficiency of all of them:

Concept	L/D _e	Design gross weight, lb
Quadrotor, turboshaft	4.9	3 678
Quiet single main rotor, turboshaft	5.4	3 951
Side-by-side, electric	7.2	5 547
Lift + cruise, electric	7.9	9 482
Lift + cruise, turbo-electric	8.5	7 271
Tiltwing, turbo-electric	8.6	6 584

The tiltwing is best in cruise efficiency *and* lighter than either lift-plus-cruise concept. A framework that predicted "best in cruise implies heaviest" would be refuted by this row. The framework here predicts no such thing: it says the tiltwing satisfies most of the escape condition, because the same propulsors serve hover and cruise and nothing is left exposed, and that it pays instead for the mechanism that rotates them. That is precisely the trade Section 3.6 finds when it sizes a tilting layout itself, and it is why no claim of superiority over the tilting family is made anywhere in this paper.

The comparison of Section 3.6 shows the same pattern on a different set of architectures: of the three sized there, the tilting layout has the best cruise lift-to-drag ratio — 13.44 against 12.00 — and is nonetheless twenty percent heavier than the tail-sitter, because it carries a tilt mechanism that the tail-sitter does not. Best in f_2 , worse in f_1 . **That comparison is an illustration and not evidence, and the distinction matters here.** Its tilting layout is given a cruise-drag multiplier of 1.00 — that is, its mechanism is credited as aerodynamically free — precisely to make the f_2 advantage as large as the architecture could possibly claim. A comparison whose inputs were chosen by the present authors cannot corroborate the present authors' framework. **The evidential weight rests on work done by others**, whose numbers were produced for other purposes and are not ours to choose; Section 3.6 shows what the framework looks like when applied, not that it is right.

A second independent check exists, and it is on aircraft that were built rather than sized. Bacchini and Cestino compare three flying eVTOLs — one per architecture — on five parameters [21]:

	E-Hang 184 (multirotor)	Cora (lift + cruise)	Lilium (vectored thrust)
Disc loading, N m ⁻²	440	880	7500
Total hover time, min	20.5	16.5	12.1
Cruise speed, km h ⁻¹	100	180	252
Practical range, km	42	107	203

The ranking reverses completely between the hover rows and the cruise rows. The architecture best in hover is worst in cruise and the architecture best in cruise is worst in hover, with the lift-plus-cruise layout between them on every line — which is what it means for the currencies to be separate rather than three names for one quantity. Those authors

also state two of the three transfers in their own words, without any framework to state them in. Of the lift-plus-cruise aircraft: its "parasitic drag caused by the pylons and vertical thrust propellers increases the power required in cruise" — that is f_2 . And of the vectored-thrust aircraft, whose cruise efficiency is the best of the three: its hover "is so power demanding that it requires batteries with higher specific power" than those assumed, so that "the aerodynamic advantages of this configuration are balanced by higher demands on the batteries and on the power electronics" — that is f_2 bought and f_3 paid, named as an exchange by an author who was not looking for one.

None of these comparisons validates the framework. All are external consistency checks: the NASA study was carried out for other purposes and its numbers were not chosen to suit the argument here, and a framework that predicted the opposite ordering would be in difficulty against them. The Bacchini and Cestino comparison is weaker as evidence in one specific way and stronger in another: weaker because its three aircraft differ in mass, mission and technical maturity as well as in architecture, so it does not isolate the mechanism the way a controlled sizing study does; stronger because they exist, and their numbers are not the output of anyone's sizing loop. Corroboration of this kind raises confidence that the three charges are separable in practice; it does not establish that they are the only three, and nothing short of a broad survey of sized architectures could.

What the framework does not claim. It does not predict the magnitude of any bill for an architecture that has not been sized; the fractions above must be computed or measured case by case. Nor does it claim that these are the only architectural costs a VTOL aircraft carries: control authority, thermal management, transition hardware, reliability and certification are all real and none of them is one of these three. What it provides is narrower and, because it is narrower, defensible — that these three recurring charges follow from the duty-cycle mismatch of Section 2.1, that they are the currencies in which the architectural remedies surveyed in Section 2.5 trade against one another, and that there is a stateable condition under which none of the three is charged.

S6.2 A comparative sizing of three architectures

Sections 3.2 to 3.5 argue that a particular configuration declines a particular trade. That argument is made against the general statement of the tax in Section 2, not against any competing aircraft, and an argument of that shape has a known weakness: it can be right about the mechanism and still be wrong about the outcome, because a rival architecture may pay the bills and recover more than it pays. The claim is therefore tested here by sizing the same mission three ways.

Method. One set of equations is used for all three, and they are the equations of Section 2.12 — closed-loop mass, hover power from momentum theory, and a Breguet-type range:

$$\begin{aligned} \mathrm{MTOW} &= \frac{m_{\text{payload}}}{1 - f_{\text{empty}} - f_{\text{fuel}}}, \\ \text{P}_{\text{hover}} &= \frac{W^{3/2}}{\eta_h \sqrt{2\rho A}}, \quad \text{R} = \frac{f_{\text{fuel}}}{E^* \eta_{\text{chain}}} \frac{L}{D} \end{aligned}$$

The propulsion-chain mass is not a fixed fraction. It is split into a part proportional to take-off mass and a part proportional to installed power, because a fixed fraction would make the third bill invisible by construction. Installed power depends on take-off mass and take-off mass depends on installed power, so the system is closed by fixed-point iteration.

Calibration. Every coefficient is back-solved from the light reference design of Section 3.7 rather than assumed: a hover figure of merit of 0.599 from 10.9 kW at 50 kg, a cruise propulsive efficiency of 0.721 from 1.7 kW at L/D 12, an engine rating margin of 1.53, and a power-independent propulsion fraction of 0.108 given an assumed 1.0 kW kg⁻¹ for a small engine and generator. The model must then reproduce the design it was calibrated from, and it does — take-off mass, propulsion fraction, engine rating, lift-to-drag ratio, range and hover power all within 0.1 percent. Run at the heavy design point without retuning, it predicts 1 037 kg against 1 000 kg and 1 813 km against 1 814 km; the one term that does not carry across is the engine rating margin, discussed in Supplementary S5.13.

What differs between the architectures. Mission, wing loading, disc loading, fuel fraction, structural fraction, avionics fraction and energy chain are held identical. Only three things change, and each is either a measurement quoted elsewhere in this paper or an openly swept parameter:

	Cruise L/D multiplier	Architecture-specific mass	Source
A — tail-sitter	1 / 1.12	—	Section 3.3, tip-frame drag
B — lift + cruise	13 / 17	second propulsion group, swept	Section 2.3, wind tunnel
C — tilt	1.00	tilt mechanism, swept	assumed, not measured

Result. With the same buffered series-hybrid power system given to all three — which neutralises the third bill, and does so against the proposed configuration:

	Empty fraction	MTOW	L/D	Hover power	Range
A — tail-sitter	0.580	50.0 kg	12.00	10.9 kW	1 600 km
B — lift + cruise	0.689	86.0 kg	10.28	18.7 kW	1 370 km
C — tilt	0.624	60.3 kg	13.44	13.1 kW	1 792 km

Against lift-plus-cruise the result is unambiguous and it is driven by measurement: the same mission closes at seventy-two percent higher take-off mass and fourteen percent lower range, and the drag term behind it is a wind-tunnel result, not an assumption. Giving the lift-plus-cruise layout the additional structural fraction that distributed lift is generally held to require makes its mass worse still — 117 kg at four additional points of structure — without changing its range at all, so the comparison as tabulated is generous to it rather than the reverse.

Against tilt the table above goes the other way, and both reasons must be stated plainly. The tilting layout closes lighter than lift-plus-cruise and cruises twelve percent further than the proposed configuration. Neither part of that outcome is a finding. The first reason is the multiplier of 1.00, which credits the tilting layout with paying no cruise

drag at all for its nacelles, pivots, actuators and hover-pitched blades. The second is subtler and belongs to the sizing rule rather than to any architecture.

Range in the equation above contains the fuel fraction and not the fuel mass. Holding the fraction fixed across architectures — the natural choice, and the one the table uses — lets the heavier aircraft carry proportionally more fuel, which removes the mass bill from the range column entirely. The general form is

$$R = \frac{E^* \eta_{\text{chain}}}{g} \frac{L}{D} \frac{m_{\text{fuel}}}{\text{MTOW}}$$

so that a fixed fraction makes range independent of take-off mass, a fixed fuel *mass* makes it inversely proportional to take-off mass, and a fixed take-off mass with a fixed payload leaves fuel as the residual.

These are three different questions, and which one is the right question depends on what is being procured: a mission, a fuel load, or a vehicle class. The mission stated in Section 3.7 — 13 kg of payload over roughly 1600 km, with take-off mass free to close where it will — is closest to the first, which is also the only rule under which the tilting layout leads, and leads only because its mechanism was credited as aerodynamically free. Reporting that column on its own would restate the credit as a conclusion. **All three are therefore given equal standing, and no result from this section should be quoted without the rule it was computed under.** The tilting layout keeps its zero cruise-drag credit throughout:

Range relative to the tail-sitter	Fixed fuel fraction	Fixed fuel mass	Fixed MTOW and payload
B — lift + cruise	−14.4 %	−36.5 %	−72.6 %
C — tilt	+12.0 %	+0.2 %	−19.1 %

Against lift-plus-cruise the conclusion is the same under every rule and grows more emphatic as the rule tightens. Against tilt it is not: the twelve percent advantage becomes a tie when the two aircraft carry the same fuel, and a nineteen percent deficit when they are the same take-off mass carrying the same payload — because at 50 kg the tilting layout's empty fraction leaves 0.116 for fuel where the proposed configuration leaves 0.160. Sweeping the cruise-drag multiplier across all three rules gives the full picture:

Tilt cruise-drag multiplier	Fixed fuel fraction	Fixed fuel mass	Fixed MTOW and payload
1.00	+12.0 %	+0.2 %	−19.1 %
0.96	+7.5 %	−4.3 %	−23.6 %
0.92	+3.0 %	−8.9 %	−28.2 %
0.88	−1.4 %	−13.4 %	−32.7 %

Of the twelve cells, the tilting layout leads in three, all of them in the first column and all of them requiring its mechanism to be aerodynamically free. Under the fixed fraction the sign changes at a multiplier of approximately 0.89, which is to two decimal places the

penalty the proposed configuration charges itself for its own tip frames, $1/1.12 = 0.893$. **The comparison therefore supports a conditional and not a ranking: under a fixed fuel-fraction rule, and with the tilt mechanism assumed aerodynamically free, the tilting layout cruises further; under equal fuel mass or equal take-off mass that advantage disappears.** No claim of superiority over the tilting family is made here in either direction.

Two smaller points belong with that disclosure. The cruise propulsive efficiency is also shared with the tilting layout, which is generous, since a blade pitched for hover is not the blade one would choose for cruise. **An earlier version of this section claimed that range does not contain propulsive efficiency, and that is wrong:** the propeller is the last link of the η_{chain} that appears in the range equation above, at 0.80 of the overall 0.176. The generosity therefore falls on both columns. On mass, 60.3 kg becomes 62.7 kg at a fifteen percent efficiency penalty; on range, the tilting layout's fixed-fraction advantage of twelve percent falls to roughly **minus five percent**, since range is linear in η_{chain} . The correction moves the comparison against the tilting layout rather than for it, which is why it is recorded here rather than left as a rounding matter: the one column in which that architecture led is the column the correction removes it from. The sizing tables above are not recomputed on it, because the fifteen percent is an illustration and not a measurement; what the tables report is the comparison with the credit left in place, which is the generous case. And a second table, in which each architecture is given its own power system with no buffer, is not reported as a fair comparison and should not be read as one: a real lift-plus-cruise aircraft hovers on batteries rather than on an engine sized for hover, so that table is a bounding case for an unbuffered series hybrid and not a description of the architecture it is labelled with.

What this comparison does and does not support. It supports the claim that the proposed configuration avoids the mass and drag bills that a separate lift system pays, and it supports it with the paper's own measurements rather than by assertion. It does not support a claim of superiority over the tilting family under every sizing rule, and the paper does not make one; what it shows is that the tilting family's apparent advantage survives only one of the three rules, and only on an assumption that was not measured. The tilting family answers the same escape condition by a different route — the same hardware, reused, but reoriented by a mechanism — and the case for the configuration proposed here rests on reaching that reuse without the mechanism, together with its control and transition consequences, and not on out-cruising it.